

difficult times, try practicing the technique of deep breathing. Just fill your lungs as much as you can and exhale slowly. After about 5 to 10 breaths, you will usually find that you have calmed down and your inner balance has been restored.

- Have a notepad, organizer, tablet, or any other digital application accessible at all times.
- Make daily, weekly, and monthly schedules that show exam times, study sessions, and any other regular commitments. Plan leisure time around these commitments rather than the other way around (Figure 2-6).
- Identify times during the day when you are energetic and times when you want or need to relax. Plan your schedule accordingly.
- Reward yourself with a special treat or activity for work well done and efficient time management.
- Do not neglect physical activity. Remember that exercise and recreation stimulate clear thinking.
- Schedule at least one block of free time each day. This free time will be your hedge against events that happen unexpectedly, such as car trouble, child-care problems, helping a friend in need, or other unforeseen circumstances.
- Understand the value of to-do lists for the day and the week. These lists help prioritize tasks and activities, a key element to organizing time efficiently (Figure 2-7).
- Make effective time management a habit.

figure 2-6

Schedules help keep track of your commitments—exam times, study sessions, work shifts, and so on.



figure 2-7

To-do lists prioritize tasks and activities.



Employ Successful Learning Tools

After reading this section, you will be able to:

LO6 Demonstrate good study habits.

Having a successful career as a professional barber begins by employing key learning tools while in school. To realize the greatest benefits education can provide, commit yourself to doing the following:

- Attend all classes.
- Arrive for class early.
- Have all necessary materials ready.
- Listen attentively to your instructor.
- Take notes.
- Highlight important points.
- Pay close attention during summary and review sessions.
- When something is not clear, ask for clarification. If you are still unsure, ask again for assistance (Figure 2-8).

figure 2-8

Do not be afraid to ask for help.



Everyone learns in their own personal and unique ways. Your study skills should utilize the methods or tools that will help you absorb and retain new information. Learning all of the study skills available will help you find what works best for you and make valuable use of your time.

REPETITION

Whether you repeat information in your head, say it out loud, write it down, or practice it hands-on, **repetition** helps your short-term memory secure a firmer grasp on the information, making the information easier to retrieve when you need it.

ORGANIZATION

Organization can be used to process new information for both short-term and long-term memory use. If a new topic seems particularly overwhelming, categorize the information into smaller segments. For example, rather than trying to learn about all the layers of skin at once, study and learn about one skin division at a time.

MNEMONICS

A **mnemonic** (NEW-mon-ick) is a device that helps you remember or recall information; they can be word associations, acronyms, songs or rhymes, or any other form of memory trigger that helps you to recall information.

Word Associations

To promote better long-term memory, try to associate new information with prior knowledge by using word association techniques. For example, the outermost layer of the epidermis is the stratum corneum. It is also known as the horny layer because it consists of tightly packed cells that are similar to an animal's horn or hoof. The outer cells are continually shed and replaced with new cells from the underlying layers of the epidermis. In this example, *corn* in *corneum* rhymes with *horn*, providing an easy way to remember both technical and alternate terminology.

Using word associations can be helpful for remembering most information but be sure to create associations that mean something to you. Meaningful associations will assist you in actually learning the material and make it easier for you to retrieve from your long-term memory when you need it.

Acronyms

Acronyms are created by using the first letters in a series of words (**Figure 2-9**). For example, the functions of the skin can be remembered by using the word SHAPES—sensation, heat regulation, absorption, protection, excretion, and secretion. SHAPES is a particularly good acronym because the skin also gives shape to the body.

Songs or Rhymes

Songs or rhymes do not have to be complicated. Something as simple as “keep the air and the hair moving when blowdrying” to prevent burning the client's scalp or “rock 'n roll rodding creates a spiral perm” to illustrate a rodding technique can be effective reminders during application procedures.

figure 2-9
Acronyms connect related ideas and make them more memorable.



VISUAL STUDY SKILLS

For some, the need to visually lay out thoughts, plans, and ideas is key to retaining information. For these learners, the more visual study skills of mind mapping and note taking can be effective learning tools.

Mind Mapping

Mind mapping (MYND MAP-ing) is a method you can use to create a visual representation of your thoughts, ideas, or class notes (**Figure 2-10**). Here are some basic guidelines for mind mapping a topic:

- Write the main topic or problem in the center of a piece of paper.
- Think about the topic and allow your ideas to flow.
- Write down key words or ideas that come to mind.
- Use lines to connect the key words to the main topic.
- Expand on the key words by creating new connections to additional thoughts or information.
- Use colors and/or symbols to highlight important information.

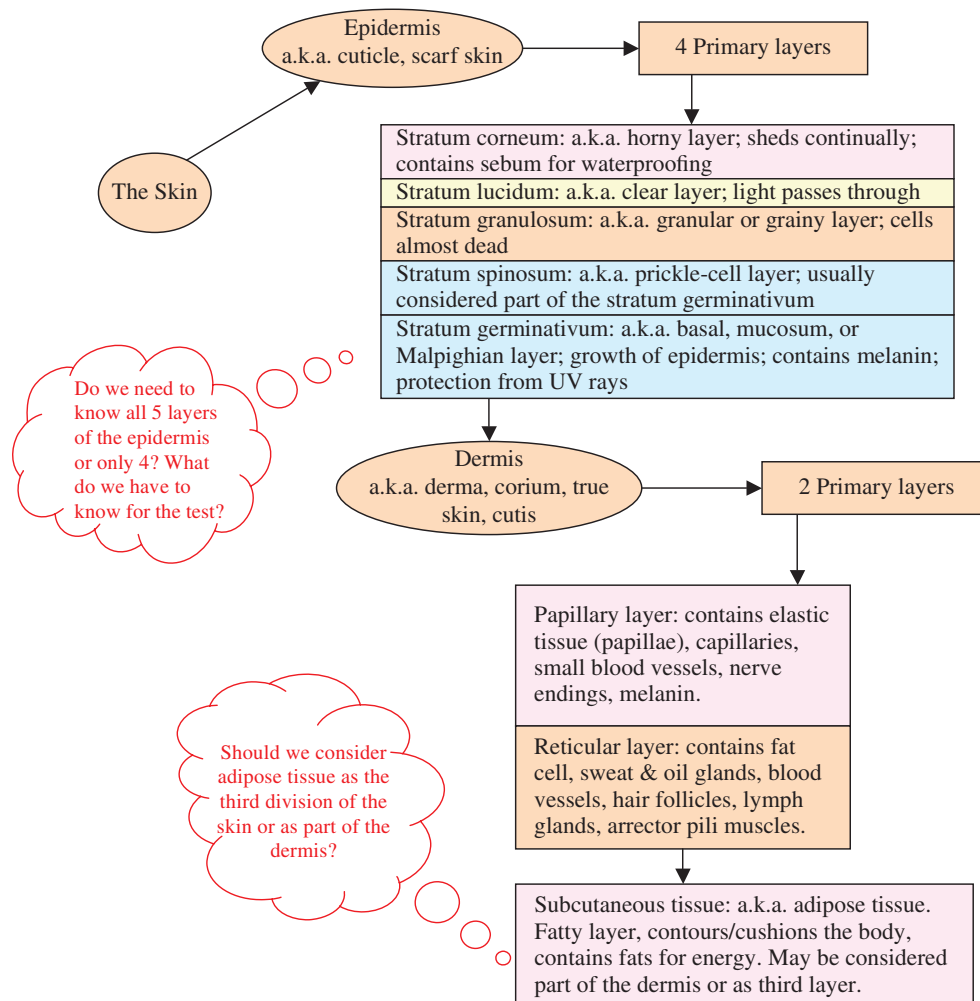


figure 2-10
Mind mapping skin divisions and layers.

Taking Notes

Note taking is a valuable skill to develop for studying and working in the business world. Effective note takers listen carefully and pay attention to verbal cues that alert them to important information. When taking notes, listen and watch for these cues:

- The speaker emphasizes words or phrases.
- You hear a directive, such as “You need to know” or “You might want to remember.”
- The speaker uses definitive words such as *main* or *first* that provide the importance or the order of something.
- If it gets written on the board, it is usually important.

Although most of us develop our own way of taking notes, here are some general tips to get you started. Remember, you need to take notes in a way that makes sense to you and that will help you make the most of the process.

- Develop or select a note-taking template that works best for the way you like to organize new information.
- Date and number the note pages for easier organization when you need to review them.
- Listen carefully to lesson introductions and summaries.
- Use key words or phrases to identify main points.
- Use complete and accurate sentences when the instructor says “You need to know ...” or “This is important” or when technical definitions are used.
- Use marks or symbols to emphasize important words, definitions, and so on.
- Use symbols, pictures, and diagrams to create visual reminders or to illustrate a concept.
- Use colored pens or highlighters to emphasize important points.

Remember that every effort you make to follow through on your education is an investment in your future. The progress you make with your learning will increase your confidence. In fact, when you have mastered a range of information and techniques, your self-esteem will soar right along with your grades.

Practice Ethical Standards

After reading this section, you will be able to:

LO 7 Define ethics.

Ethics (ETH-iks) are the moral principles by which we live and work. In a shop setting, ethical standards should guide your conduct with clients and fellow employees. When your actions are respectful, courteous, and helpful, you are behaving in an ethical manner.

✓ HERE'S A TIP

Always remember to review your notes after class to transfer new information from short-term memory to long-term memory.

✓ HERE'S A TIP

After becoming a licensed professional, seek continuing education opportunities. Never stop learning! The barbering industry is constantly changing and growing; there are always new trends, techniques, products, and information. Reading magazines, joining industry associations, attending trade shows, and enrolling in advanced educational classes are all ways to continue learning. For access to continuing education opportunities, visit MiladyPro.com.

Practice ethical behavior in the barbershop by employing these five professional actions:

- Provide skilled and competent services
- Be honest, courteous, and sincere
- Avoid sharing clients' private matters with others—even your closest friends
- Participate in continuing education and staying on track with new information, techniques, and skills
- Give clients accurate information about treatments and products

PROFESSIONAL ETHICS

To be an ethical person, you need to embody the following qualities:

- **Self-care.** To be helpful to others, it is essential to take care of yourself. Try the Self-Care Test to assess how you are doing (Figure 2-11).
- **Integrity.** Maintain your integrity by aligning your behavior and actions to your values. For example, recommending products that clients do not really need is unethical behavior. On the other hand, if you feel that a client would benefit from certain products and additional services, it would be unethical not to give the client that information.

figure 2-11
Self-Care Test.

The Self-Care Test

Some people know intuitively when they need to stop, take a break, or even take a day off. Other people forget when to eat. You can judge how well you take care of yourself by noting how you feel physically, emotionally, and mentally. Here are some questions to ask yourself to see how you rate on the self-care scale.

1. Do you wait until you are exhausted before you stop working?
2. Do you forget to eat nutritious food and substitute junk food on the fly?
3. Do you say you will exercise and then put off starting a program?
4. Do you have poor sleep habits?
5. Are you constantly nagging yourself about not being good enough?
6. Are your relationships with people filled with conflict?
7. When you think about the future are you unclear about the direction you will take?
8. Do you spend most of your spare time watching TV?
9. Have you been told you are too stressed and yet you ignore these concerns?
10. Do you waste time and then get angry with yourself?

Score 5 points for each yes. A score of 0–15 says that you take pretty good care of yourself, but you would be wise to examine those questions you answered yes to. A score of 15–30 indicates that you need to rethink your priorities. A score of 30–50 is a strong statement that you are neglecting yourself and may be headed for high stress and burnout. Reviewing the suggestions in this chapter will help you get back on track.

- **Discretion.** Do not share your personal issues with clients. Likewise, never breach confidentiality by repeating personal information that clients have shared with you.
- **Communication.** Your responsibility to behave ethically extends to your communications with customers and coworkers. Be aware of what you say and how you say it. Also, be conscious of your nonverbal communications, such as facial expressions or body language, which are just as important as verbal communication.

Develop a Positive Personality and Attitude

After reading this section, you will be able to:

- LO8** List the characteristics of a healthy, positive attitude.



DID YOU KNOW?

The Whole Person

An individual's personality is the sum of their characteristics, attitudes, and behavioral traits. Attitude improvement is a process that continues throughout life. In both your professional and personal life, a pleasing attitude gains more associates, clients, and friends. You will know you have a pleasing attitude when you are able to see the good in difficult situations. People enjoy the company of individuals who can put a positive “spin” on things.

Barbers interact with people from all walks of life—every day, all day. It is useful, therefore, to have a sense of how different personality traits work together. Refer regularly to the following characteristics of a healthy, positive attitude:

- **Diplomacy.** Being assertive is good because it helps people understand your position. However, it is a short step from being assertive to being aggressive or even bullying. Take your attitude temperature to see how well you practice the art of diplomacy. Diplomacy—also known as tact—is the ability to deliver truthful, even sometimes critical or difficult, messages in a kind way.
- **Pleasing tone of voice.** The tone of your voice is a personality trait, but if your natural voice is harsh or if you tend to mumble, you can consciously improve by speaking more softly or more clearly. Another technique is to smile when speaking if it is appropriate. Smiling will help the tone of your voice, so practice it when speaking in person and when talking on the phone.
- **Emotional stability.** Learning how to handle a confrontation and how to share your feelings in a professional manner is important to building emotional stability and control (**Figure 2-12**).
- **Sensitivity.** Being sensitive means being compassionate and responsive to other people.
- **Values and goals.** Values and goals guide our behavior and give us direction.
- **Receptivity.** Be interested in other people and responsive to their opinions, feelings, and ideas. Receptivity involves taking the time to really listen to others. Also, be open minded and willing to work with all personality types.
- **Effective communication skills.** Commit to practicing effective communication through active listening, nonverbal, and verbal skills.

figure 2-12

Emotional stability plays a key part in successfully handling confrontations.



© Dmytro Zinkevych/Shutterstock.com.

REVIEW QUESTIONS

- 1 What principles contribute to personal and professional success?
- 2 How do you create a mission statement? Give an example.
- 3 How do you go about setting long-term and short-term goals?
- 4 What are some of the most effective ways to manage time?
- 5 How do you describe good study habits?
- 6 What are some learning tools you should use while studying?
- 7 What is the definition of ethics?
- 8 What are the characteristics of a healthy, positive attitude?

CHAPTER GLOSSARY

ethics (ETH-iks)	p. 34	the moral principles by which we live and work
game plan	p. 26	the conscious act of planning your life, instead of just letting things happen
goal setting	p. 28	the identification of long-term and short-term goals that helps you decide what you want out of your life
mind mapping (MYND MAP-ing)	p. 33	a graphic representation of an idea or problem that helps organize one's thoughts
mission statement (MISH-uhn STATE-ment)	p. 27	a statement that establishes the purpose and values for which an individual or institution lives and works by. It provides a sense of direction by defining guiding principles and clarifying goals, as well as how an organization operates
mnemonic (NEW-mon-ick)	p. 32	any memorization device that helps a person recall information
organization	p. 32	a method used to store new information for short-term and long-term memory
perfectionism (PUR-fek-shun-izm)	p. 26	an unhealthy compulsion to do things perfectly
prioritize (PRIH-or-uh-tize)	p. 30	to make a list of tasks that need to be done in the order of most-to-least important
procrastination (PRO-crass-tin-aye-shun)	p. 26	putting off until tomorrow what you can do today
repetition	p. 32	repeatedly saying, writing, or otherwise reviewing new information until it is learned
self-actualization	p. 25	fulfilling one's full potential

3



PROFESSIONAL
IMAGE

LEARNING OBJECTIVES

After completing this chapter, you will be able to:

CHAPTER OUTLINE

INTRODUCTION

WHY STUDY PROFESSIONAL IMAGE?

APPLY HEALTHFUL HABITS IN YOUR DAILY ROUTINE

FOLLOW IMAGE-BUILDING BASICS

EMPLOY PROPER ERGONOMICS TO PROTECT YOUR BODY

PRACTICE EFFECTIVE HUMAN RELATIONS AND COMMUNICATION SKILLS

LO¹

Name four important personal hygiene habits.

LO²

Explain the concept of dressing for success.

LO³

Practice ergonomically correct movement, postures, and principles.

LO⁴

Demonstrate an understanding of human relations and communication skills.

figure 3-1
Put your best professional image forward.



Introduction

Do you believe that first impressions are important? First impressions are often the gateway to obtaining a job interview, getting new customers, or building a professional image. Making a positive impact is essential when working in the business of personal style. Barbers are often held to higher image standards by clients because they view barbers as image experts. For this reason alone, it is vital to look and act your absolute best when in public because the image you project sets the standard for clients and peers in the workplace (**Figure 3-1**).

There are many factors that help create a professional image. However, for better or worse, how a person looks is often the first clue to determining if that person has what it takes to do the job. From there, professional behavior, a positive attitude, team camaraderie, proper ergonomics, and good communication skills all contribute to your total image.

why study PROFESSIONAL IMAGE?

Barbers should study and have a thorough understanding of professional image because:

- Clients rely on barbers to look good and be well groomed. Having a professional image helps build trust with clients and leads to repeat business.
- Working in a barbershop whose culture complements your image standards and goals is important for career growth and achievements.
- Understanding ergonomics can help prevent health issues associated with poor working habits and help professionals to stay gainfully employed.
- There are consequences to not maintaining a professional image, including loss of clients, a poor reputation, and loss of income.

Apply Healthful Habits in Your Daily Routine

After reading this section, you will be able to:

LO 1 Name four important personal hygiene habits.

In accordance with the general concept of a profession that helps others look their best, barbers should strive to present their own best image as well. An important part of this representation is the barber's personal health and physical appearance. To achieve success in this area, it is helpful to follow a set of guidelines that help maintain a healthy body and a healthy mind.

HYGIENE

Basic hygienic practices such as showering or bathing should never be omitted from daily personal care practices. **Personal hygiene** (pur-sun-AL HY-jene) is the daily maintenance of cleanliness by practicing good healthful habits. When working as a barber, you will be in close proximity to clients. One weak moment of drinking coffee right before performing a service, or wearing something that needs laundering because you did not plan ahead, could be disastrous. Rather than telling you that you smell offensive, most clients will simply not return and may tell others about their bad experience.

The following personal habits result in good hygiene throughout the day:

- Wash your hands throughout the day as required, including at the beginning of each service (**Figure 3-2**).
- Perform self-checks, and wash or freshen under the arms as needed.
- Brush and floss your teeth, and use mouthwash or breath mints throughout the day.
- If you smoke cigarettes, *do not* smoke during work hours. Many clients find the lingering smell of smoke offensive. If you smoke during your lunch break, brush your teeth, use mouthwash, and wash your hands afterward.

figure 3-2

Hand washing is the first step in maintaining good hygiene.



REST AND RELAXATION

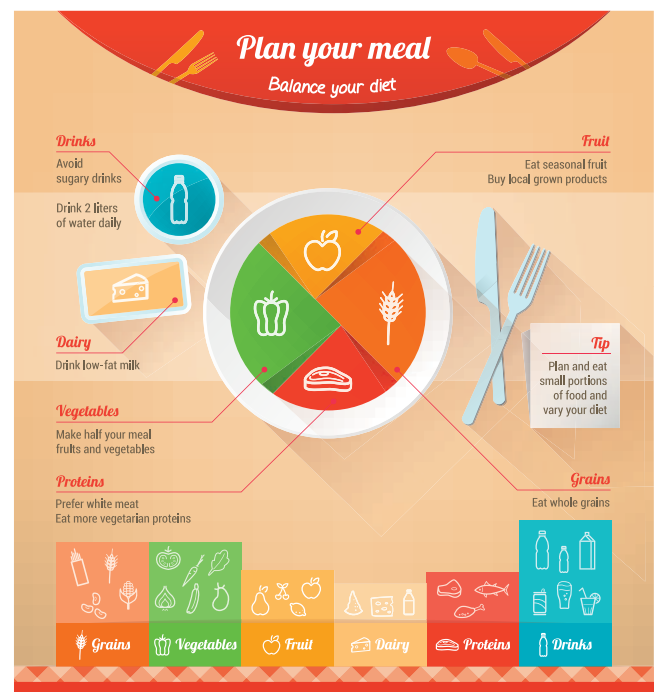
Adequate sleep is essential for good health because without it you cannot function efficiently. The body needs to be allowed to recover from the fatigue of the day's activities and should be replenished with a good night's sleep. The amount of sleep needed to feel refreshed varies from person to person. Some people function well with 6 hours of sleep while others need 8 hours. An average of 7 or 8 hours of sleep each night is recommended by medical professionals. Relaxation is also important as a change of pace from day-to-day routines. Going to a movie or museum, reading a book, watching television, playing sports, or dancing are just some ways to escape and unwind. When you return to work, you should feel refreshed and eager to attend to your duties.

NUTRITION

What you eat affects your health, appearance, personality, and performance on the job. A balanced diet should include foods containing a variety of important vitamins and minerals (**Figure 3-3**). Drink plenty of water daily. Try to avoid sugar, salt, caffeine, and fatty or highly refined and processed foods.

figure 3-3

Meal planning can help maintain a balanced diet.




MILADYPRO LEARN MORE!

Optional info on **professionalism** can be found at MiladyPro.com; keyword: FutureBarberPro.

figure 3-4
Strive for harmony in your life.



EXERCISE

Exercise and recreation in the form of walking, dancing, sports, and gym activities tend to develop the muscles and help keep the body fit. Regular physical activity benefits the body by improving blood circulation, oxygen supply, and proper organ function.

STRESS MANAGEMENT AND A HEALTHY LIFESTYLE

Stress is the inability to cope with a real or imagined threat that results in a series of mental and physical responses or adaptations. The way in which individuals perform under stressful situations depends on their temperament, physical health, and coping skills. Practice stress management through a combination of rest, relaxation, exercise, and daily routines that provide you with time to calm the body and its systems. Live a life of moderation in which work and other activities are balanced so that you can achieve a sense of harmony in your life (Figure 3-4).

© GaudiLab/Shutterstock.com.

Follow Image-Building Basics

After reading this section, you will be able to:

LO 2 Explain the concept of dressing for success.


CAUTION

Often barbershops have a no-fragrance policy for staff members because a significant number of people are sensitive or allergic to a variety of chemicals, including perfume oils. Whether or not your barbershop has a no-fragrance policy, cologne and perfume should not be worn at work.

Being well groomed advertises a barber’s commitment to the barbering industry. Consider yourself a representative of the profession and make sure to follow personal grooming habits and to practice professional behaviors. If you present a poised and attractive image, your clients will have confidence in you as a professional.

PERSONAL GROOMING

Many barbershop owners and managers view appearance and personality as being just as important as technical knowledge and skills. **Personal grooming** is the process of caring for parts of the body and maintaining an overall polished look. How you dress and how you take care of your hair, skin, and nails reflect your personal grooming habits.

Dress for Success

While working, your wardrobe selection should express a professional image that is consistent with the image of the barbershop (Figure 3-5). Your **professional image** (pruh-FESH-un-ul IM-aje) is the impression you project through both your outward appearance and your conduct in the workplace. Your clothes must be pressed and clean—not simply free of the dirt that you can see, but stain free, an objective that is sometimes difficult to achieve in a barbershop environment. Be mindful about spills and drips when using chemicals, and avoid leaning on counters in the work area—particularly in

the dispensary or storage room. Although many shop owners do not require their barbers to wear standard uniforms, they may have a specific dress code for the shop. For example, some may require the wearing of a barber's jacket, some a smock, others a tie, or specific colors. These requirements are examples of a barbershop's culture and how barbers can dress for success.

When shopping for work clothes, visualize how you would look and function in them while performing services. Is the image you will present one that is acceptable to your clients? It is important to consider body type and shape when choosing what to wear to work. Select clothing that looks flattering and be sure to look in the mirror from various angles before leaving for work. Avoid both excessively baggy and overly tight clothes, which can get in the way of your work.

Dressing for success does not mean that you have to be someone that you are not. Everyone has a unique personality and it is okay to let your personality speak through your clothing, shoes, and accessories. Just remember to tune in to the barbershop's culture (Figure 3-6).

While it is important to always follow the employer's dress code, here are some universal wardrobe guidelines:

- Wear clothing that is clean, fresh, and current with fashion.
- Choose clothing that is functional, comfortable, and stylish.
- Accessorize your outfits, but make sure that your jewelry does not make noise while you work, dangle in the client's face, or get caught in equipment.
- Wristwatches, while a good idea to help stay on schedule, should be waterproof.
- Wear shoes that are supportive, polished, and in good repair. Ill-fitting shoes are not the best choices to wear when a lot of standing is required.

Hair Maintenance

Complement your professional wardrobe with an up-to-date hairstyle. It is important to keep your haircut and style fresh (Figure 3-7). To accomplish this, schedule time for a hair appointment, and possibly barter with other barbers for hair care services. Even the pros deserve to be pampered. Remember, you are a representative of the profession!

Facial grooming for male barbers is also of utmost importance. If you maintain a beard or mustache, be sure it is neat and trimmed; for the clean-shaven barber, shave daily and avoid undue stubble. Your personal facial grooming can further testify to your barbering skill, especially in this heyday of facial hair.

Skin Care and Makeup

Just like a smile, proper skin care can help promote a professional image. Having and maintaining healthy skin is an ongoing process. Develop a skin care regimen that works best with your skin type (Figure 3-8). Use protective products on your skin such as sunscreens; if skin problems occur, seek professional advice. If used, makeup should enhance facial features, so take your time and apply makeup prior to arriving to work or school. Applying makeup at your station is unprofessional and advertises that you have poor time management skills.

figure 3-5

Your work wardrobe should complement the image of your barbershop.

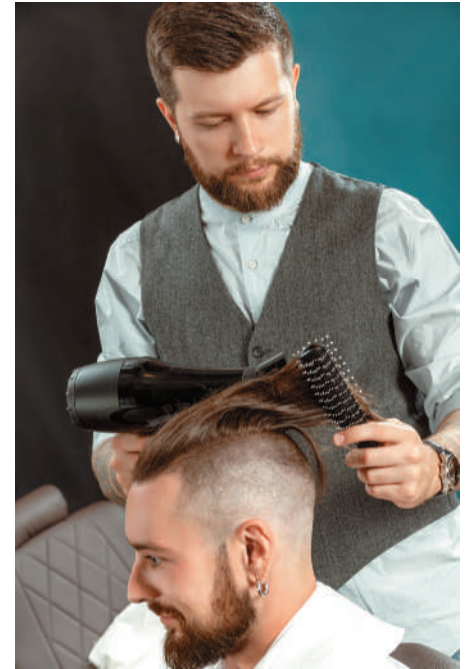


figure 3-6

Express yourself and your shop.



figure 3-7

Make time to keep your own hair up to date.



figure 3-8

Taking care of your skin is important for both your health and your image.



Nail Care

Barbering professionals have the privilege of using their hands to make a living, but often neglect their own care. Manicures are a great way to relax hands and thoroughly clean nails. Determine a nail length that suits your personal style, and maintain nails as needed. Broken nails or chipped nail polish may happen occasionally but should not be a regular occurrence.

Employ Proper Ergonomics to Protect Your Body

After reading this section, you will be able to:

LO 3 Practice ergonomically correct movement, postures, and principles.

CAUTION

Not only can wearing inappropriate shoes at work be uncomfortable, it could be dangerous. Flip-flops and open-toed shoes are not safe to wear around electrical tools and sharp implements.

Your **physical presentation** (FIZ-ih-kuhl pres-ent-TAY-shun) involves your posture and the way you walk and move. Good posture conveys an image of confidence and can prevent fatigue and many other physical problems. Sitting or standing improperly can put a great deal of stress on your neck, shoulders, back, and legs. Having good posture allows you to get through your day feeling good and doing your best work.

POSTURE

Working as a barber, you will spend most of your time in a standing position. Raising your arms to work at the top of the client's head or bending over at the shampoo bowl are repetitive motions that can create physical stress in the hand, wrist, arm, shoulder, and lower-back areas. When such stresses occur, physical or movement therapies may be required to maintain proper body alignment.

Practice the following guidelines for maintaining a more stress-free standing posture behind the chair:

- Keep your head up and chin parallel to the floor.
- Keep your neck elongated and balanced directly above the shoulders.
- Lift your upper body so that your chest is up and out—do not slouch.
- Hold your shoulders level and relaxed.
- Stand with your spine straight.

Just as there is a mechanically correct posture for standing, there is also a correct sitting posture. Use the following guidelines to learn to sit correctly in a balanced position:

- Keep your hips level and horizontal, not tilted forward or backward.
- Flex your knees slightly and position them over your feet.
- Lower your body smoothly into a chair, keeping your back straight.
- Keep the soles of your feet on the floor directly under your knees.
- Have the seat of the chair even with your knees. This position will allow the upper and lower legs to form a 90-degree angle at the knees.
- Rest the weight of your torso on the thighbones, not on the end of the spine.
- Keep your torso erect.
- When sitting at a desk, make sure it is at the correct height so that the upper and lower parts of your arm form a right angle when you are writing.

BODY MOVEMENT

Your muscles and bones work together as a musculoskeletal system, allowing you to walk, raise your arms, and use your fingers. **Ergonomics** (UR-go-nom-icks) is the science of designing the workplace as well as its equipment and tools to make specific body movements more comfortable, efficient, and safe.

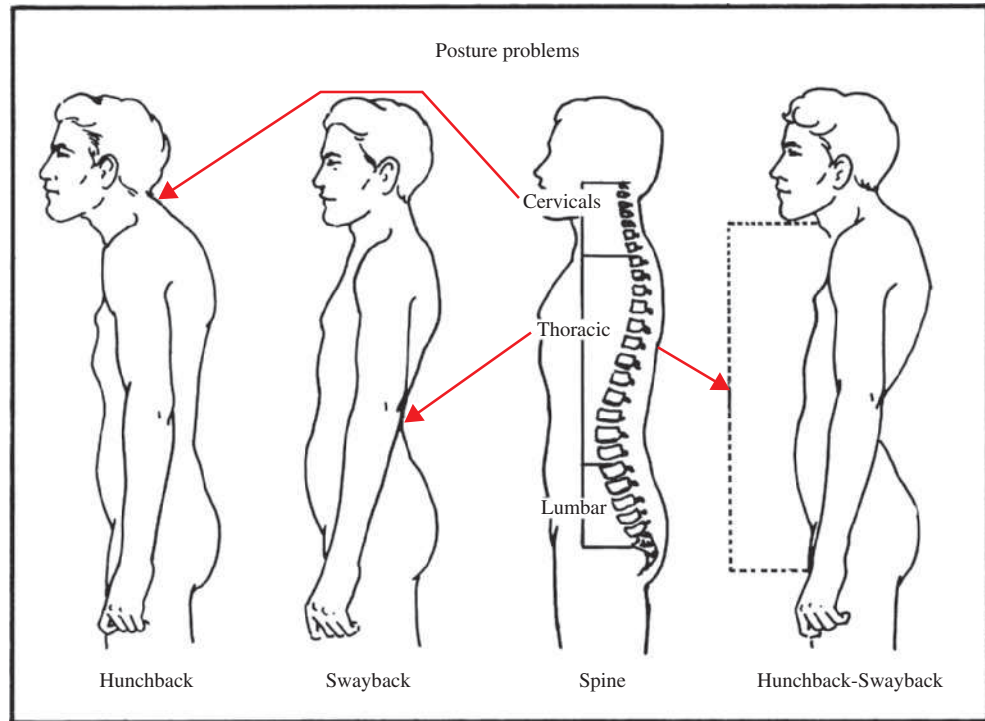
For example, a hydraulic chair can be raised or lowered to accommodate barbers of different heights, allowing each barber to service clients without bending over too far. Certain shears are designed to eliminate hand fatigue when cutting hair because repetitive movements are of particular concern.

Each year, hundreds of barbering professionals report musculoskeletal disorders, including carpal tunnel syndrome (a wrist injury) and back injuries (**Figure 3-9**). Barbers may have to stand or sit all day and perform repetitive movements, which makes them susceptible to problems of the hands, wrists, shoulders, neck, back, feet, and legs.

Prevention is the key to avoiding problems. An awareness of your posture and movements, coupled with good work habits, proper tools, and equipment, will enhance your health and comfort.

Ergonomics is important to your ability to work and your body's wellness. Repetitive motions have a cumulative effect on the muscles and joints.

figure 3-9
Common posture problems.



To avoid problems and work more effectively, practice some of the following suggestions:

- Do not grip or squeeze tools and implements too tightly.
- Do not bend the wrist up or down constantly when cutting hair or using a blowdryer (**Figure 3-10**).
- Try to position your arms at less than a 60-degree angle when holding your arms away from your body while working.
- Avoid bending forward and/or twisting your body to get closer to your client.
- Adjust the height of the chair so that the client's head is at a comfortable working level.
- Tilt the client's head as necessary for better access during hair services.

To avoid ergonomic-related injuries, consider these further guidelines:

- Keep your wrists in a straight or neutral position as much as possible (**Figure 3-11**).
- Use ergonomically designed implements.
- Keep your back and neck straight.
- Stand on an antifatigue mat.
- When standing to cut hair, position your legs hip-width apart, bend your knees slightly, and align your pelvic area with your abdomen.

figure 3-10
Improper haircutting position.



Counter the negative impact of repetitive motions or long periods spent in one position by stretching and walking around at intervals. Always put your well-being first.

Activity

Practice these quick exercises to help you relieve stress from repetitive movements or from standing or sitting in one position for too long.

For Wrists

1. Stand up straight.
2. Raise both of your arms straight out.
3. Bend your wrists so your fingers point upward and hold for 5 seconds.
4. Hold your wrists steady and turn your hands, so your fingers face the floor and hold for 5 seconds.
5. Repeat the cycle five times.

For Fingers

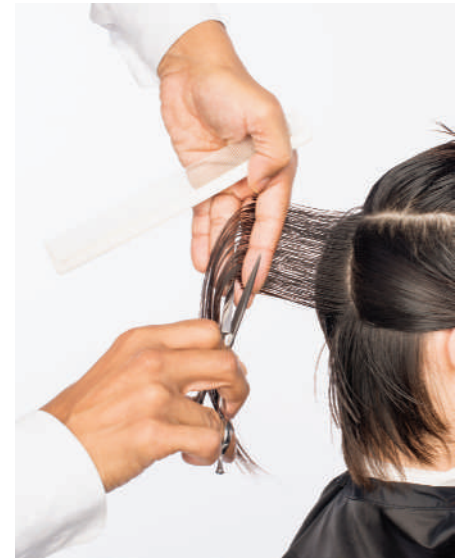
1. Get a ball the size of a tennis ball or a tension ball.
2. Grip it tightly for a count of five. Release.
3. Repeat five times.

For Shoulders

1. Stand up straight and shrug your shoulders upward.
2. Roll your shoulders back and hold for a count of five.
3. Reverse direction and roll your shoulders forward for a count of five.
4. Repeat five times.

figure 3-11

Correct wrist and hand position for haircutting.



Practice Effective Human Relations and Communication Skills

After reading this section, you will be able to:

LO 4 Demonstrate an understanding of human relations and communication skills.

Human relations describes the interactions and relationships between two or more people. Effective human relations skills help you build rapport with clients and coworkers. Rapport involves establishing a close and empathetic relationship that fosters agreement and harmony between individuals. Your professional attitude is expressed through your self-esteem, confidence, and the respect you show others (**Figure 3-12**). Good habits and practices acquired during your education lay the foundation for a successful career in barbering.

figure 3-12

Treat clients with respect, empathy, and confidence.



The following guidelines for good human relations will help you gain confidence, deal courteously with others, and become a successful professional:

- Always greet clients by name, using a pleasant tone of voice. Address clients by their last name, as in “Mr. Jones” or “Mrs. Smith,” unless the client prefers first names or it is customary to use first names in the barbershop.
- Be alert to the client’s mood. Some clients prefer quiet and relaxation; others like to talk. Be a good listener and confine the conversation to the client’s needs. Never gossip or tell off-color stories.
- Topics of conversation should be carefully chosen. Friendly relations are achieved through pleasant conversations. Let the client guide the topic of conversation. In a business setting, it is best to avoid such controversial topics as religion and politics, personal problems, or issues relating to other people. Never discuss other clients or coworkers, and always maintain an ethical standard of confidentiality. Never discuss personal wages, tips, rent, or tax information.
- Make a good impression by looking the part of the successful barber and by speaking and acting in a professional manner at all times.
- Cultivate self-confidence and project a pleasing personality.
- Show interest in the client’s personal preferences and give the client undivided attention.
- Use tact when dealing with problems you may encounter.
- Deal with all disputes and differences in private. Take care of all problems promptly and to the client’s satisfaction.
- Be capable and efficient.
- Be punctual. Arrive at work on time and keep appointments on schedule. Plan your day so that you have time to take breaks and enjoy a healthy lunch.

In addition to basic human relations skills, barbering students need to cultivate other aspects of a professional image that can impact their success in the profession. Some desirable qualities for effective client relations are the following:

figure 3-13
Recognize and control your emotions.



- **Talking less, listening more.** When you practice good listening skills, you are fully attentive to what the other person is saying. If there is something that you do not understand, ask questions to gain understanding.
- **Emotional control.** Learn to control your emotions. Try to respond rather than to react. Do not reveal negative emotions such as anger, envy, and dislike through gestures, facial expressions, or conversation. An even-tempered person is usually treated with respect (**Figure 3-13**).
- **Positive approach.** Be pleasant and gracious. Be ready with a smile of greeting and a word of welcome for each client and coworker. A good sense of humor is also important in maintaining a positive attitude.

© g-stockstudio/Shutterstock.com.

- **Good manners.** Good manners reflect your thoughtfulness toward others. Treating others with respect, exercising care of other people's property, being tolerant and understanding of their shortcomings and efforts, and being considerate of those with whom you work all express good manners. Courtesy is one of the most important keys to a successful career.
- **Mannerisms.** Gum chewing and nervous habits such as tapping your foot or playing with your hair detract from the effectiveness of your image. Yawning, coughing, and sneezing should be concealed with your hand in front of your mouth. Control negative body language—sarcastic or disapproving facial grimaces, for example. Exhibiting pleasant mannerisms and attractive gestures and actions should be your goal at all times.

EFFECTIVE COMMUNICATION SKILLS

Communication is one of the barber's most important human relations skills. Communication includes listening skills, tone, speech, and conversational ability—all of which are necessary to forming satisfying relationships with customers and coworkers.

Effective communication (uh-FEK-tive COM-yun-ik-ay-shun) is the act of successfully sharing information between two people (or groups of people) so that the information is understood (**Figure 3-14**). In the barbershop, effective communication requires sending and receiving messages in order to establish a relationship with the client. The following steps can be used to help you determine your clients' service expectations.

- **Organize your thoughts.** What question or information do you want your client to understand? For example, a client with medium-length hair that covers the top part of the ears says he wants a trim. You will need to determine just what the client's definition of a trim is and may ask if he wants the tops of his ears covered or not.
- **Clarify.** The next step is to clarify what the client is telling you. In the preceding scenario, let's say the client's response to the question is that he wants his hair "over the ears." This answer still does not provide you with the information you need to proceed with the haircut. Why? Because you now have to clarify what the client means by "over the ears." Does it mean covering the tops of the ears or does it mean above and/or around the ears?
- **Repeat.** Once you have an understanding of the client's definition of "over the ears" or any other description used, repeat to the client your interpretation of what you think he told you. This communication step will provide you with further clarification and the opportunity to make needed changes—in short, to reach an understanding of the client's expectations.

figure 3-14

Communication is key to understanding the client's needs and wants.



- Making contacts and networking
- Meeting and greeting clients
- Understanding a client's service needs, likes, dislikes, desires, and expectations
- Self-promotion and building a clientele
- Selling services and products
- Telephone conversations and appointment bookings
- Conversation and interaction with the shop staff, clients, and vendors

Finally, a frequent form of communication that can easily be misunderstood, yet has serious impact upon your business, is online communication (**Figure 3-15**). Establishing a professional online image is an essential image-building attribute, and not one to be taken lightly. Social-media websites, including photo-sharing sites, can quickly diminish a person's reputation if media etiquette is neglected.

- Manage your personal pages and walls.
- Use social media to communicate with peers and clients.
- Post helpful content.

- Use profane language.
- Participate in or entertain arguments online.
- Post nude or embarrassing photographs.
- Forward spam.

© best works/Shutterstock.com.

REVIEW QUESTIONS

- 1 What are the five basic requirements for personal and professional health?
- 2 What are four important personal hygiene habits?
- 3 What is the best way to ensure you are dressed for success?
- 4 What are five ways you can avoid ergonomic-related injuries?
- 5 What is *rapport*?
- 6 What are five desirable qualities for effective client relations?
- 7 What are the three communication steps that help determine a client's service expectations?
- 8 What are four things not to do when it comes to social media?

CHAPTER GLOSSARY

effective communication (uh-FEK-tive COM-yun-ik-ay-shun)	p. 49	the act of successfully sharing information between two people (or groups of people) so that the information is understood
ergonomics (UR-go-nom-icks)	p. 45	the science of designing the workplace as well as its equipment and tools to make specific body movements more comfortable, efficient, and safe
human relations	p. 47	the interactions and relationships between two or more people
personal grooming	p. 42	the process of caring for parts of the body and maintaining an overall polished look
personal hygiene (pur-sun-AL HY-jene)	p. 41	the daily maintenance of cleanliness by practicing good healthful habits
physical presentation (FIZ-ih-kuhl pres-ent-TAY-shun)	p. 44	your posture, as well as the way you walk and move
professional image (pruh-FESH-un-ul IM-aje)	p. 42	the impression you project through both your outward appearance and your conduct in the workplace

PART 2 GENERAL SCIENCES

4 INFECTION CONTROL:
PRINCIPLES AND
PRACTICES

5 IMPLEMENTS, TOOLS,
AND EQUIPMENT

6 GENERAL ANATOMY AND
PHYSIOLOGY

7 BASICS OF CHEMISTRY

8 BASICS OF
ELECTRICITY

9 THE SKIN – STRUCTURES,
DISORDERS, AND
DISEASES

10 PROPERTIES AND
DISORDERS OF THE
HAIR AND SCALP





4



INFECTION CONTROL: PRINCIPLES AND PRACTICES

CHAPTER OUTLINE

PUBLISHER'S NOTE

INTRODUCTION

WHY STUDY INFECTION CONTROL:
PRINCIPLES AND PRACTICES?

MEET THE CURRENT
REGULATIONS FOR HEALTH
AND SAFETY

UNDERSTANDING THE PRINCIPLES
OF INFECTION

PREVENT THE SPREAD OF DISEASE

FOLLOW STANDARD
PRECAUTIONS TO PROTECT YOU
AND YOUR CLIENTS

FOLLOW SAFE WORK PRACTICES
AND SAFETY PRECAUTIONS

LIST YOUR PROFESSIONAL
RESPONSIBILITIES

PROCEDURES 4-1 TO 4-4

LEARNING OBJECTIVES

After completing this chapter, you will be able to:

LO1

Discuss federal and state agencies that regulate the practice of barbering.

LO2

List the types and classifications of bacteria.

LO3

Define bloodborne pathogens and explain how they are transmitted.

LO4

Explain the differences between cleaning, disinfecting, and sterilizing.

LO5

Identify types of disinfectants and antiseptics appropriate for use in barbershops.

LO6

Discuss Standard Precautions and explain procedures for handling an exposure incident.

LO7

Discuss safe work practices that help prevent accidents and injuries.

LO8

List your responsibilities as a professional barber.

Publisher's Note

In the previous edition of this text, the term *sanitation*, also known as *sanitizing*, was used interchangeably to mean, “clean” or “cleaning.” You will also find that many commercially available products used in the cleaning and disinfecting process continue to use the words *sanitize* and *sanitizing*. However, the publisher’s goal is to clearly define these terms in this book, in textbooks across the industry, and within the glossary.

Before we discuss infection control and safe work practices, the terms *cleaning*, *sanitizing*, *disinfecting*, and *sterilizing* need to be properly differentiated, because of the following factors:

- There is much confusion about and misuse of the terms *cleaning*, *sanitizing*, *disinfecting*, and *sterilizing* within the professional beauty and barbering industry. In an effort to do what we can to clarify these critical terms, Milady opted to consistently use the term *cleaning*, instead of using *cleaning* in one sentence and *sanitizing* in another sentence.
- Professionals in the healthcare and scientific communities (of disease prevention and epidemiology) and associations, such as the Association for Professionals in Infection Control and Epidemiology, generally do not use the terms interchangeably. Instead, it is more common for infection control professionals to use the term *cleaning*. Infection control professionals consider **sanitation** (san-ih-TAY-shun) a layperson’s term or a product marketing term (as in *hand sanitizers*).
- The term *cleaning* is defined as a mechanical process (scrubbing) using soap and water or detergent and water to remove all visible dirt, debris, and many disease-causing germs. Cleaning also removes invisible debris that interferes with disinfection. Cleaning is required as a step before disinfecting.
- The term *sanitizing* is defined as a chemical process for reducing the number of disease-causing germs on cleaned surfaces to a safe level.
- The term *disinfecting* is defined as a chemical process for use with nonporous items that uses specific products to destroy harmful organisms (except bacterial spores) on implements and environmental surfaces.

Introduction

State barber boards and other regulatory agencies require that infection control measures and safe work practices be applied while serving the public. Infection control measures help minimize the spread of contagious diseases and skin infections that are caused by the transmission of infectious material from one individual to another. Since transmission can also occur when using contaminated implements, tools, or equipment, the performance of effective infection control procedures must be a top priority in the barbershop.

Safe work practices require that implements, tools, and equipment be used safely and that you be aware of situations that can cause accidents in the barbershop. This chapter provides some helpful guidelines to minimize potential risks and accidents.

It is your responsibility as a professional barber to use proper and effective infection control methods that help safeguard your health and the health of your clients. You are also responsible for employing safe work practices to help prevent accidents and injuries from occurring in the workplace.

why study

INFECTION CONTROL: PRINCIPLES AND PRACTICES?

Barbers should study and have a thorough understanding of infection control: principles and practices because:

- It is important to know about the pathogens you and your clients may be exposed to in the barbershop and their modes of transmission.
- Understanding and practicing proper infection control within the laws and rules will help safeguard your health, the health of your clients, and your business.
- It is important to respect the procedures for handling chemicals used in cleaning and disinfecting by reading labels and following manufacturers' instructions to keep you and your clients safe.
- Practicing safety precautions on a daily basis protects your clients and your license.
- A responsible professional barber is conscientious about infection control and safety.

Meet the Current Regulations for Health and Safety

After reading this section, you will be able to:

LO 1 Discuss federal and state agencies that regulate the practice of barbering.

Many federal and state agencies regulate the practice of barbering. Federal agencies set guidelines for the manufacturing, sale, and use of equipment and chemical ingredients. These guidelines also monitor safety in the workplace and place limits on the types of services you can perform in a barber or styling shop. State agencies regulate licensing, enforcement, and your conduct when you are working in the shop.

FEDERAL AGENCIES

Occupational Safety and Health Administration

The Occupational Safety and Health Administration (OSHA) was created as part of the U.S. Department of Labor to regulate and enforce safety and health standards to protect employees in the workplace. Regulating



WEB RESOURCES

You can find a list of disinfectants approved by the Environmental Protection Agency (EPA) by going to the EPA's website, at <http://www.epa.gov> and entering a search on the home page for EPA-registered disinfectants. Disinfectants are not listed as "hospital grade" but instead are listed based on the pathogens they are effective against. Products on list D meet the criteria of most states for hospital disinfectants; products on list E meet the criteria of a tuberculocidal in those states where that is required.

employee exposure to potentially toxic substances and informing employees about the possible hazards of materials used in the workplace are key points of the Occupational Safety and Health Act of 1970. This regulation created the Hazard Communication Standard (HCS), which requires that chemical manufacturers and importers assess and communicate the potential hazards associated with their products. The Material Safety Data Sheet (MSDS) was a result of the HCS. In 2012, along with representatives from most of the nations in the United Nations, OSHA agreed to comply with the Globally Harmonized System of Classification and Labeling of Chemicals System (GHS). This initiative was designed to create label standards to be used around the globe and includes the use of specific pictograms to indicate possible safety concerns; the initiative also includes the adoption of a 16-category, standard-format Safety Data Sheet (SDS) to replace the MSDS. The HCS Act in 1983 gave workers the *right to know*, but the new GHS gives workers the *right to understand*.

The standards set by OSHA are important to barbers because of the products they use daily. OSHA standards address issues relating to the handling, mixing, storing, and disposing of products; general safety in the workplace; and your right to know about any potentially hazardous ingredients contained in the products and how to avoid these hazards. Barbers should view OSHA as an agency designed to ensure a safe workplace for all U.S. workers.



DID YOU KNOW?

The term *hospital grade* is a term that is not used by the EPA. The EPA does not grade disinfectants; a product either is approved by the EPA for use as a hospital disinfectant or is not.

Safety Data Sheets (SDS) Replace Material Safety Data Sheets (MSDS)

Both federal and state laws require that manufacturers supply a **Safety Data Sheet** (previously known as *Material Safety Data Sheet* [muh-TEE-ree-uhl SAYF-tee DAY-tuh SHEET]) for all chemical products manufactured and sold (**Figure 4-1**). All SDSs will be organized identically and will contain the following 16 categories of information:

1. **Identification.** Product identifier, manufacturer or distributor with contact information (including emergency phone number), recommended use of the product, and restrictions on use are displayed.
2. **Hazard identification.** All hazards of using the chemical are detailed.
3. **Composition/information on ingredients.** Information on chemical ingredients is included.
4. **First-aid measures.** Important symptoms/effects (acute and delayed) and required treatment are included.
5. **Firefighting measures.** Suitable fire-extinguishing techniques and equipment and the chemical hazards of fire are listed.
6. **Accidental release measures.** Emergency procedures, protective equipment, and proper methods of containment and clean-up are listed.
7. **Handling and storage.** Precautions for safe handling and storage, including incompatibilities, are listed.
8. **Exposure controls/personal protection.** OSHA's permissible exposure limits and personal protective equipment are listed.
9. **Physical and chemical properties.** The chemical's characteristics are listed.

figure 4-1
Safety Data Sheet (SDS).



© Travis Klein/Shutterstock.com.

10. **Stability and reactivity.** Chemical stability and possibility of hazardous reactions are detailed.
11. **Toxicology information.** Routes of exposure, related symptoms, and acute and chronic effects are included.
12. **Ecological information.** Effects on wastewater and environment are detailed.
13. **Disposal consideration.** Proper disposal techniques and disposal restrictions are detailed.
14. **Transport information.** Restrictions on transportation are included.
15. **Regulatory information.** Agencies responsible for the regulation of the product are listed.
16. **Revision date.** Original date of document and any revisions are listed.

Source: Adapted from United States Department of Labor. (n.d.) *OSHA QuickCard—Hazard Communication Safety Data Sheets*. Retrieved from <http://www.OSHA.gov>.

In addition, pictograms that are internationally recognized are used (Figure 4-2) to ensure that information is being communicated in easily recognizable formats. When necessary, the SDS can be sent to a medical facility so that a doctor can better assess and treat the patient. Not having SDS information available poses a health risk to anyone exposed to hazardous materials and violates federal and state regulations. OSHA and state

HCS Pictograms and Hazards

Health Hazard  • Carcinogen • Mutagenicity • Reproductive Toxicity • Respiratory Sensitizer • Target Organ Toxicity • Aspiration Toxicity	Flame  • Flammables • Pyrophorics • Self-Heating • Emits Flammable Gas • Self-Reactives • Organic Peroxides	Exclamation Mark  • Irritant (skin and eye) • Skin Sensitizer • Acute Toxicity (harmful) • Narcotic Effects • Respiratory Tract Irritant • Hazardous to Ozone Layer (non-mandatory)
Gas Cylinder  • Gases Under Pressure	Corrosion  • Skin Corrosion/ Burns • Eye Damage • Corrosive to Metals	Exploding Bomb  • Explosives • Self-Reactives • Organic Peroxides
Flame Over Circle  • Oxidizers	Environment (Non-Mandatory)  • Aquatic Toxicity	Skull and Crossbones  • Acute Toxicity (fatal or toxic)

figure 4-2
Pictograms used on SDSs.

regulatory agencies require that SDSs be kept available in the shop for all products used or sold. Both OSHA and state board inspectors can issue fines to establishments not having SDSs readily available.

Federal and state laws require barbers to obtain SDSs from the chemical product manufacturers and/or distributors for each professional product they use. SDSs often can be downloaded from the product manufacturer's or the distributor's website. All workers in the barbershop must read the information included on each SDS and verify that they have read it by adding their signatures to a sign-off sheet for the product. All states require that sign-off sheets must be immediately available to state and federal inspectors upon request.

Environmental Protection Agency (EPA)

The EPA registers all types of disinfectants sold and used in the United States. **Disinfectants** (dis-in-FEK-tents) are chemical products that destroy most bacteria (excluding spores), fungi, and viruses on surfaces. The two types that are used in the barbershop are hospital disinfectants and tuberculocidal disinfectants.

Hospital disinfectants (HOS-pih-tal dis-in-FEK-tents) are effective for cleaning blood and body fluids from nonporous surfaces. By **nonporous** (nahn-POHW-rus), we mean an item made or constructed of a material that has no pores or openings and that cannot absorb liquids. Hospital disinfectants control the spread of **disease** (DIZ-eez). Disease is defined as an abnormal condition of all or part of the body, its systems, or its organs that make the body incapable of carrying on normal functions. The most commonly used disinfectants are the following:

- Quaternary ammonium compounds, commonly known as *quats*, are products made of quaternary ammonium cations and are designed for disinfection of nonporous surfaces. They are appropriate for use in noncritical (noninvasive) environments and are effective against most pathogens of concern in the barbering environment.
- **Tuberculocidal disinfectants** (tuh-bur-kyoo-LOH-syd-ahl dis-in-FEK-tents), often referred to as *phenolics*, are proven to kill the bacterium that causes **tuberculosis** (tuh-bur-kyoo-LOH-sus), in addition to other pathogens destroyed through the use of hospital disinfectants. Tuberculosis is a disease caused by a bacterium that is transmitted through coughing or sneezing. It is passed through inhalation only and is not transmitted by the hands or picked up on surfaces.

Tuberculocidal disinfectants are one kind of hospital disinfectant; however, because these disinfectants are effective against the pathogen does not mean that you should automatically reach for them. Some of these products can be harmful to tools and equipment and they require special methods of disposal. Check the rules of your state to be sure that the product you choose complies with state requirements.

Most pathogens of concern in the barbershop are adequately destroyed by standard EPA-registered disinfectants and do not require tuberculocidal disinfectants; however, some states *do* require tuberculocidal disinfectants, so always check the rules of your state to be sure the product complies with state requirements. There are also some states that require tuberculocidal



figure 4-3
All products and containers need to be labeled.

disinfectants only when blood contamination has occurred. In those states, you should keep a tuberculocidal disinfectant readily available, but you should use it only when required.

It is against federal law to use any disinfecting product in a way contrary to the use indicated on its label. Before a manufacturer can sell a product for disinfecting surfaces, tools, implements, or equipment, they must obtain an EPA registration number (shown following “EPA Reg. No.” near the manufacturer’s name) that certifies that the disinfectant, when used correctly, will be effective against the pathogens listed on the label. For example, clipper disinfectants must be approved by the EPA for use with clippers in specific environments (such as a barbershop) or the manufacturer would be breaking federal law by marketing them as clipper disinfectants to the barber market. This also means that if you do not follow the label instructions for mixing, contact time, and the type of surface the disinfecting product can be used on, you are not complying with federal law (**Figure 4-3**). If there were an injury-related lawsuit, you could be held responsible.

STATE REGULATORY AGENCIES

State regulatory agencies exist to protect barbers and their customers’ health and safety while they receive services. State regulatory agencies include licensing agencies, state barber boards, commissions, and health departments. Regulatory agencies require that everyone working on consumers in a barbershop follow specific procedures. Enforcement of the rules through inspections and investigations of consumer complaints is also part of an agency’s responsibility. An agency can issue penalties against both the shop owner’s and the barber’s licenses. Penalties vary and include warnings, fines, probation, and suspension or revocation of licenses. It is vital that you understand and follow the laws and rules of your state at all times. Your professional reputation, your license, and your clients’ safety depend on it.



CAUTION

Remember, barbers are not allowed to treat or recommend treatments for infections, diseases, or abnormal conditions. Customers with such problems should be referred to their physicians.

LAWS AND RULES—WHAT IS THE DIFFERENCE?

Laws are written by both federal and state legislatures that determine the scope of practice (what each license allows the holder to do) and establish guidelines for regulatory agencies to make rules. Laws are also called *statutes*.

Rules and regulations are more specific than laws. The regulatory agency or the state board writes the rules and determines how the law must be applied. Rules establish specific standards of conduct and can be changed or updated frequently. It is the barber's responsibility to be aware of any changes to the rules and regulations and to comply with them. Ignorance of the law is not an acceptable reason or excuse for noncompliance.

Understanding the Principles of Infection

After reading this section, you will be able to:

- LO 2 List the types and classifications of bacteria.
- LO 3 Define bloodborne pathogens and explain how they are transmitted.

Being a barber professional is not only rewarding, but it is also a great responsibility. One careless action could cause injury or **infection** (in-FEK-shun), which is the invasion of body tissues by disease-causing pathogens. If your actions result in a client's injury or infection, you could lose your license or ruin the shop's reputation. Fortunately, preventing the spread of infections is easy when you know proper procedures and follow them at all times. Prevention begins and ends with *you*.

Effective infection control in the barbershop also influences the professional image of the establishment. A client's first impression of the shop begins the moment they open the door; so a clean environment should extend beyond each barber's immediate work area. All of the sights, sounds, smells, and general ambience of the barbershop meld together to form this first impression regardless of how many times a client has previously visited the shop. In essence, this first impression happens over and over again because clients do not see you or the shop daily. A clean and orderly barbershop helps build client confidence and trust that continuous care is being taken to provide a safe and sanitary environment in which to receive their personal services.

INFECTION CONTROL

Infection control (in-FEK-shun CON-trol) refers to the methods used to eliminate or reduce the transmission of **infectious** (in-FEK-shus) organisms. Barbers must understand and remember the following four types of potentially harmful organisms:

- Bacteria
- Fungi
- Viruses
- Parasites

Under certain conditions, many of these organisms can cause infectious diseases. An **infectious disease** (in-FEK-shus DIZ-eez) is caused by pathogenic (harmful) organisms that enter the body. An infectious disease, however, may or may not be spread from one person to another depending on the organism and its method of transmission.

In this chapter, you will learn how to properly clean and disinfect the tools and equipment you use so they are safe for you and your customers.

Cleaning (KLEEN-ing) is a mechanical process (scrubbing) using soap and water or detergent and water to remove all visible dirt, debris, and many disease-causing germs from tools, implements, and equipment.

The process of **disinfection** (dis-in-FEK-shun) involves the use of a chemical to destroy most, but not necessarily all, harmful organisms on environmental surfaces. Disinfection is not effective against **bacterial spores** (bak-TEER-ee-ul SPOORZ), which are bacteria capable of producing a protective coating that allows them to withstand very harsh environments and to shed the coating when conditions become more favorable to them.

Cleaning and disinfecting procedures are designed to prevent the spread of infection and disease. At minimum, disinfectants used in barbershops must be

- **bactericidal** (BAK-TEER-uh-syd-uhl), capable of destroying bacteria;
- **virucidal** (VI-ra-syd-uhl), capable of destroying viruses;
- **fungicidal** (FUN-ji-syd-uhl), capable of destroying molds and fungi.

Remember, in some states, disinfectants must also be effective against tuberculosis (tuberculocidal). Check your state board rules and regulations for compliance information.

Be sure to mix and use these disinfectants according to the instructions on the labels so they are safe and effective.

Contaminated tools and equipment can spread infections between customers if proper disinfection steps are not taken after every service. You have a professional and legal obligation to protect your clients from harm by using proper infection control procedures. If clients are infected or harmed because you did not perform infection control procedures correctly, you may be found legally responsible for their injuries or infections.

BACTERIA

Bacteria (bak-TEER-ee-ah) (singular: bacterium [bak-TEER-ee-uhm]) are single-celled microorganisms that have both plant and animal characteristics. A **microorganism** (my-kroh-OR-gah-niz-um) is any organism of microscopic or submicroscopic size. Some bacteria are harmful and some are harmless. Bacteria can exist almost anywhere: on skin, in water, in the air, in decayed matter, on environmental surfaces, in body secretions, on clothing, or under the free edge of nails. Bacteria are so small they can only be seen with a microscope.



HERE'S A TIP

This is a quick checklist for a disinfectant label to be in compliance with state and federal regulations:

- The EPA registration number is prominently displayed.
- The label indicates “hospital grade” and/or if it is bactericidal, virucidal, and fungicidal on label.
- The label shows the disinfectant’s efficacy list—What does it kill? Look for HIV (human immunodeficiency virus), HBV (hepatitis B virus), and MRSA (methicillin-resistant staphylococcus aureus) at the very least.

Types of Bacteria

There are thousands of different kinds of bacteria, which fall into two primary types: pathogenic and nonpathogenic. Most bacteria are **nonpathogenic** (nahn-path-uh-JEN-ik); in other words, they are harmless organisms that may perform useful functions. They are safe to come in contact with since they do not cause disease or harm. For example, nonpathogenic organisms are used to make yogurt, cheese, and some medicines. In the human body, nonpathogenic bacteria help the body break down food, protect against infection, and stimulate the immune system.

Pathogenic (path-uh-JEN-ik) bacteria are harmful microorganisms that can cause disease or infection in humans when they invade the body. Shops and schools must maintain strict standards for cleaning and disinfecting at all times to prevent the spread of pathogenic microorganisms. It is crucial that students learn proper infection control practices while in school to ensure that they understand the importance of following them throughout their career. **Table 4-1** presents terms and definitions related to pathogens.

table 4-1
CAUSES OF DISEASE

Term	Definition
Bacteria	One-celled microorganisms having both plant and animal characteristics. Some are harmful and some are harmless.
Direct Transmission (dy-REKT trans-MISH-uhn)	Transmission of blood or body fluids through touching (including shaking hands), kissing, coughing, sneezing, and talking.
Fungi	Singular: fungus; single-cell organisms that grow in irregular masses that include molds, mildews, and yeasts.
Indirect Transmission (IN-dih-rekt trans-MISH-uhn)	Transmission of blood or body fluids through contact with an intermediate contaminated object, such as a razor, extractor, nipper, or an environmental surface.
Infection	Invasion of body tissues by disease-causing pathogens.
Germ	Nonscientific synonym for disease-producing organisms.
Microorganism	Any organism of microscopic to submicroscopic size.
Pathogen	Harmful microorganisms that enter the body and can cause disease.
Parasite	Organisms that grow, feed, and shelter on or in another organism (referred to as the host) while contributing nothing to the survival of that organism. Parasites must have a host to survive.
Toxin	Various poisonous substances produced by some microorganisms (bacteria and viruses).
Virus	A submicroscopic particle that infects and resides in cells of biological organisms.

Classifications of Pathogenic Bacteria

Bacteria have three distinct shapes that help identify them. Pathogenic bacteria are classified as described below:

- **Cocci** (KOK-sy) are round-shaped bacteria that appear singly (alone) or in groups (**Figure 4-4**):
 - **Staphylococci** (staf-uh-loh-KOK-sy) are pus-forming bacteria that grow in clusters like bunches of grapes. They cause abscesses, pustules, and boils. Some types of staphylococci (or staph, as many call it) may not cause infections in healthy humans, although others may be deadly (**Figure 4-5**).
 - **Streptococci** (strep-toh-KOK-sy) are pus-forming bacteria arranged in curved lines resembling a string of beads. They cause infections such as strep throat and blood poisoning (**Figure 4-6**).
 - **Diplococci** (dip-lo-KOK-sy) are spherical bacteria that grow in pairs and cause diseases such as pneumonia (**Figure 4-7**).
- **Bacilli** (bah-SIL-ee) are short rod-shaped bacteria. They are the most common bacteria and produce diseases such as tetanus (lockjaw), typhoid fever, tuberculosis, and diphtheria (**Figure 4-8**).
- **Spirilla** (spy-RIL-ah) are spiral or corkscrew-shaped bacteria. They are subdivided into subgroups, such as *Treponema pallidum*, which causes syphilis, a sexually transmitted disease, and *Borrelia burgdorferi*, which causes Lyme disease (**Figure 4-9**).

Movement of Bacteria

Different bacteria move in different ways; the term **motility** (MOH-till-it-ee) refers to self-movement. Cocci rarely demonstrate self-movement and are generally transmitted in the air, in dust, or within the substance in which they settle. Bacilli and spirilla are both capable of movement and use slender, hairlike extensions called **flagella** (fluh-JEL-uh) for locomotion (moving about). You may also hear people refer to **cilia** (SIL-ee-uh) in reference to cell movement, but they are much shorter than flagella; bacteria require relatively many more cilia to ensure self-movement. Both flagella and cilia move cells, but they have different types of motion. Bacteria with flagella move in a snakelike motion while those with cilia move in a rowing-like motion.

figure 4-4
Cocci.

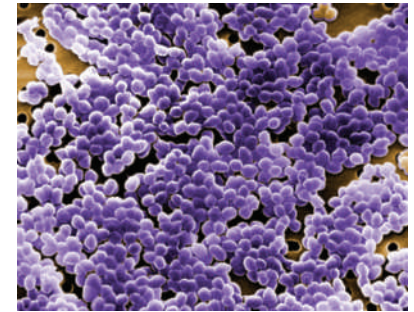


figure 4-5
Staphylococci.

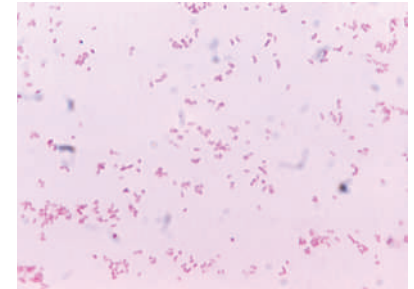


figure 4-6
Streptococci.

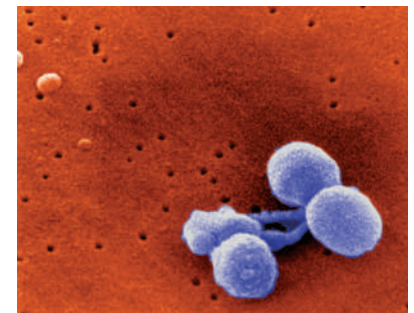
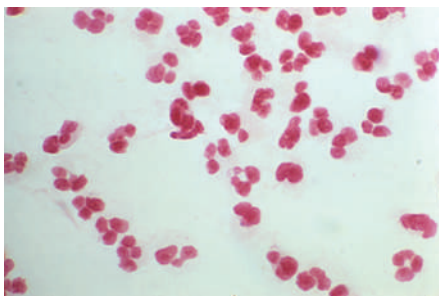


figure 4-7
Diplococci.



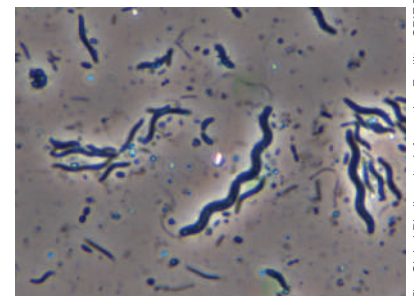
Dr. Norman Jacobs/CDC/Public Health Image Library

figure 4-8
Bacilli.



© Juan Gaermer/Shutterstock.com

figure 4-9
Spirilla.



The Global Biodiversity Information Facility. GBIF Backbone © Juan Gaermer/Shutterstock.com Dr. Norman Jacobs/CDC/Public Health Image Library Taxonomy. 2013-07-01. Accessed via <http://www.gbif.org/species/3220355> on 2013-11-22. Photo by Wolfram Adlansig

Bacterial Growth and Reproduction

When seen under a microscope, bacteria look like tiny bags. They generally consist of an outer cell wall that contains a liquid called *protoplasm*. Bacterial cells manufacture their own food through what they absorb from the surrounding environment. They give off waste products, grow, and reproduce. The life cycle of bacteria consists of two distinct phases: the active stage and the inactive stage.

Active stage. During the active stage, bacteria grow and reproduce. Bacteria multiply best in warm, dark, damp, or dirty places. When conditions are favorable, bacteria grow and reproduce. When they reach their largest size, they divide into two new cells. This division is called **binary fission** (BY-nayr-ee FISH-un). The cells that are formed as a result of binary fission are called *daughter cells* and are produced every 20 to 60 minutes, depending on the bacteria. For example, the infectious pathogen *Staphylococcus aureus* undergoes cell division every 27 to 30 minutes. When conditions become unfavorable, bacteria either die or become inactive.

Inactive stage (IN-ac-tif STAJE). Certain bacteria, such as the bacteria that cause tetanus and botulism, among others, can form spores by coating themselves with waxlike outer shells during unfavorable conditions. These spores protect the bacteria and enable them to withstand long periods of famine, dryness, and unsuitable temperatures. In this condition, spores can be blown about and are not harmed by disinfectants, heat, or cold. If favorable conditions, such as a moist environment return, the spores can change into the active form and begin to grow and reproduce (Figure 4-10).

Bacterial Infections

There can be no bacterial infection without the presence of pathogenic bacteria. Therefore, if pathogenic bacteria are eliminated, clients cannot become infected.

An **inflammation** (in-fluh-MAY-shun) is a condition in which the tissue of the body reacts to injury, irritation, or infection. An inflammation is characterized by redness, heat, pain, and/or swelling.

figure 4-10
Petri dishes with growing cultures.



© Fotofremer/Shutterstock.com.

Pus (PUS) is a fluid containing white blood cells, bacteria, and dead cells and is the by-product of the infectious process. The presence of pus is a sign of a bacterial infection. A **local infection** (LOHK-al in-FEK-shun), such as a pimple or abscess (**Figure 4-11**), is confined to a particular part of the body and appears as a lesion-containing pus. A systemic infection is an infection where the pathogen has distributed throughout the body rather than staying in one area or organ.

Staphylococci are among the most common bacteria that affect humans and are commonly found in our environment, including on our bodies, although most strains do not make us ill. Staph bacteria can be picked up on doorknobs, countertops, and other surfaces, but in the barbershop, they are more frequently spread through skin-to-skin contact (such as shaking hands) or through the use of unclean tools or implements and can be very dangerous.

Staph is responsible for food poisoning and a wide range of diseases, including toxic shock syndrome. Some types of infectious staph bacteria are highly resistant to conventional treatments such as antibiotics. An example is the staph infection called **methicillin-resistant staphylococcus aureus** (METH-eh-sill-en-ree-ZIST-ent staf-uh-loh-KOK-us OR-ee-us). Historically, MRSA occurred most frequently among persons with weakened immune systems or among people who had undergone medical procedures. Today, it has become more common in otherwise healthy people. Clients who appear completely healthy may bring this organism into the shop, where it can infect others. Some people carry the bacteria and are not even aware of their infection, but the people they infect may show more obvious symptoms. MRSA initially appears as a skin infection, presenting as pimples, rashes, or boils that can be difficult to cure. Without proper treatment, the infection becomes systemic and can have devastating consequences that can result in death. Because of these highly resistant bacterial strains, it is important to clean and disinfect all tools and implements used on customers. Also, do not perform services if the client's skin, scalp, or neck shows visible signs of abrasion or infection.

When a disease spreads from one person to another, the disease is said to be a **contagious disease** (kon-TAY-jus DIZ-eez), also known as **communicable** (kuh-MYOO-nih-kuh-bul) disease. Some of the more common contagious diseases that prevent a barber from servicing a client are the common cold, ringworm, conjunctivitis (pinkeye), and viral infections. The most common way these infections spread is through dirty hands, especially under the fingernails and in the webs between the fingers. Be sure to always wash your hands after using the restroom and before eating. Contagious diseases can also be spread by contaminated implements, cuts, infected nails, open sores, pus, mouth and nose discharges, shared drinking cups, telephone receivers, and towels. Uncovered coughing or sneezing and spitting in public also spread germs. **Table 4-2** lists terms and definitions that are important for a general understanding of disease.

VIRUSES

A **virus** (VY-rus) (plural: viruses) is a submicroscopic particle that infects and resides in the cells of a biological organism. A virus is capable of replication only through taking over the host cell's reproductive function. Viruses are so small that they can only be seen under the most sophisticated and

figure 4-11

Pimples are an example of a local infection.



table 4-2

TERMS RELATED TO DISEASE

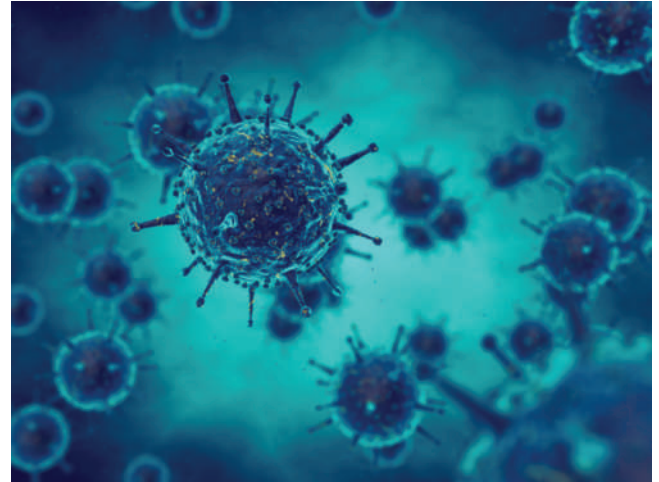
Term	Definition
Allergy	Reaction due to extreme sensitivity to certain foods, chemicals, or other normally harmless substances.
Contagious Disease	Also known as <i>communicable disease</i> ; a disease that is spread from one person to another person. Some of the more contagious diseases are the common cold, ringworm, conjunctivitis (pinkeye), viral infections, and natural nail or toe and foot infections.
Contamination (kun-tam-uh-NAY-shun)	The presence, or the reasonably anticipated presence, of blood or other potentially infectious materials on an item’s surface or visible debris or residues such as dust, hair, and skin.
Decontamination	The removal of blood or other potentially infectious materials on an item’s surface and the removal of visible debris or residue such as dust, hair, and skin.
Diagnosis (dy-ag-NOH-sis)	Determination of the nature of a disease from its symptoms and/or diagnostic tests. Federal regulations prohibit salon professionals from performing a diagnosis.
Disease	An abnormal condition of all or part of the body or its systems or organs that makes the body incapable of carrying on normal function.
Exposure Incident	Contact with non-intact (broken) skin, blood, body fluid, or other potentially infectious materials that are the result of the performance of an employee’s duties.
Infectious Disease	Disease caused by pathogenic (harmful) microorganisms that enter the body. An infectious disease may be spread from one person to another person.
Inflammation	Condition in which the body reacts to injury, irritation, or infection. An inflammation is characterized by redness, heat, pain, and swelling.
Occupational Disease	Illnesses resulting from conditions associated with employment, such as prolonged and repeated overexposure to certain products or ingredients.
Parasitic Disease	Disease caused by parasites, such as lice and mites.
Pathogenic Disease	Disease produced by organisms, including bacteria, viruses, fungi, and parasites.
Systemic Disease	Disease that affects the body as a whole, often due to under-functioning or over-functioning internal glands or organs. This disease is carried through the bloodstream or the lymphatic system.

powerful microscopes. They cause common colds and other respiratory and gastrointestinal (digestive tract) infections. Some of the viruses that plague humans are measles, mumps, chicken pox, smallpox, rabies, yellow fever, hepatitis, polio, influenza, and HIV, which causes AIDS.

One difference between viruses and bacteria is that a virus can live and reproduce only by taking over other cells and becoming part of them, while bacteria can live and reproduce on their own. Another difference is that while bacterial infections can usually be treated with specific antibiotics, viral infections cannot, and viruses are hard to kill without harming the host cells in the process. When available, vaccinations can prevent viruses from growing in the body. There are many vaccines available for viruses, but not

all viruses have vaccines. There are vaccines available for hepatitis B and varicella (the virus that causes shingles), and you should strongly consider receiving these vaccines, as well as vaccines for the seasonal flu and pneumonia (Figure 4-12).

figure 4-12
Viruses.



BIOFILMS

Biofilms are colonies of microorganisms that adhere to environmental surfaces, as well as the human body. They secrete a sticky, hard-to-penetrate, protective coating that cements them together. The biofilm grows into a complex structure, with many kinds of microbes. The sticky matrix substance holds communities together, making them very hard to pierce with antiseptics, antimicrobials, and disinfection. It keeps the body in a chronic inflammatory state that is painful and inhibits healing. One action of the biofilm community is to resist the body's defense mechanisms; we are learning that biofilms play a large role in disease and infection.

Biofilms are usually not visible and must grow very large to be seen without a microscope. Dental plaque is an example of a visible human biofilm. Numerous persistent infections, including periodontal disease, osteomyelitis, cystic fibrosis, otitis media, conjunctivitis, prostatitis, endocarditis, urinary catheter infections, contact lens and corneal infections, and infections associated with medical devices and surgical (and sterile) instruments are associated with biofilms that are not visible. Algae colonies on ponds and slime in drains are examples of environmental biofilms.

Because biofilms are hard to detect, their presence and effects seem to be underestimated. They are one of the most significant scientific discoveries of the past few decades, though we have much more to learn. Conscientiously using infection control precautions, including standard precautions, cleaning, disinfection, and sterilization, is the best method of prevention at the present time.

BLOODBORNE PATHOGENS

Disease-causing microorganisms that are carried in the body by blood or body fluids, such as hepatitis and HIV, are called **bloodborne pathogens** (BLUD-boorn PATH-o-genz). In the barbershop, the spread of bloodborne pathogens is possible during haircutting and shaving services whenever the skin is broken. Use great care to avoid cutting or damaging the customer's skin during any type of service.

Hepatitis

Hepatitis (HEP-uh-tight-is) is a bloodborne virus that causes disease and can damage the liver. In general, it is difficult to contract hepatitis. However, hepatitis is easier to contract than HIV because hepatitis can be present in all body fluids of those who are infected. In addition, unlike HIV, hepatitis can live on a surface outside the body for long periods of time. For this reason, it is vital that all surfaces that a customer comes into contact with are thoroughly cleaned and disinfected.

 **HERE'S A TIP**

There are specific things you should look for on the label of any product you use for disinfection in the barbershop. The label should always have the following:

- List of pathogens it is effective against should include HIV, HBV, and *S. aureus* (including MRSA)
- EPA registration number
- The words *bactericidal*, *virucidal*, and *fungicidal*
- Mixing and changing instructions

There are three types of hepatitis that are of concern for barbers: hepatitis A, hepatitis B, and hepatitis C. HBV is the most difficult to kill on a surface, so check the label of the disinfectant you use to be sure that the product is effective against HBV. Hepatitis B and hepatitis C are spread from person to person through blood and, less often, through other body fluids.

HIV/AIDS

The **human immunodeficiency virus** (HYOO-mun ih-MYOO-noh-di-FISH-en-see VY-rus), abbreviated HIV, is the virus that causes **acquired immune deficiency syndrome** (uh-KWY-erd ih-MYOOON di-FISH-en-see sin-droh-m), abbreviated AIDS. AIDS is a disease that breaks down the body's immune system. HIV is spread from person to person through blood and, less often, through other body fluids, such as semen and vaginal secretions. A person can be infected with HIV for many years without showing symptoms, but testing can determine whether a person is infected within 6 months after exposure to the virus. Sometimes, people who are HIV-positive have never been tested and do not know they have the potential to infect others.

HIV is spread mainly through the sharing of needles by intravenous drug users and by unprotected sexual contact. Less commonly, HIV is spread through accidents with needles in healthcare settings. The virus is less likely to enter the bloodstream through cuts and sores. Holding hands, hugging, kissing, sharing food, or using household items such as the telephone or toilet seats does not spread HIV.

If you accidentally cut a client's skin, the tool will be contaminated with whatever might be in the client's blood, including HIV. You should not continue to use the implement without cleaning and disinfecting it. Continuing to use a contaminated implement without cleaning and disinfecting it puts you and others in the barbershop at risk of infection.

FUNGI

Fungi (FUN-ji) (singular: fungus [FUN-gus]) are single-celled organisms that grow in irregular masses that include molds, mildews, and yeasts. They can produce contagious diseases, such as ringworm. **Mildew** (MIL-doo), another fungus, affects plants or grows on inanimate objects but does not cause human infections in the barbershop.

The most frequently encountered fungal infection resulting from hair services is **tinea barbae** (TIN-ee-uh BAR-bee), also known as barber's itch. Tinea barbae is a superficial fungal infection caused by a variety of dermatophytes (fungi that require keratin for growth) that commonly affects the skin. It is primarily limited to the bearded areas of the face and neck or around the scalp. This infection occurs almost exclusively in older adolescent and adult males. A person with tinea barbae may have deep, inflamed or noninflamed patches of skin on the face or the nape of the neck. Tinea barbae is similar to **tinea capitis** (TIN-ee-uh KAP-ih-tus), a fungal infection of the scalp characterized by red papules, or spots, at the opening of hair follicles.

Barbers must clean and disinfect clipper blades to avoid spreading scalp and skin infections. The risk of spreading skin and scalp infections

can be reduced by first removing all visible hair and debris from clippers followed by thorough cleaning and disinfection of the nonelectrical parts. You can use compressed air instead of a stiff brush to remove hair and debris from clippers before cleaning and disinfecting them. Turn the clippers on before applying the air to dislodge as much hair as possible during the procedure.

Always refer to the manufacturer's directions for proper cleaning and disinfecting methods and recommendations.

PARASITES

Parasites (PAYR-uh-sytz) are organisms that grow, feed, and shelter on or inside another organism (referred to as a *host*), while contributing nothing to the survival of that organism. They must have a host to survive. Parasites can live on or inside of humans and animals. They also can be found in food, on plants and trees, and in water. Humans can acquire internal parasites by eating fish or meat that has not been properly cooked. External parasites that affect humans, on or in the skin, include ticks, lice, fleas, and mites. Services should never be performed on a customer with visible signs of a parasitic infestation. Always refer the client to a physician for treatment.

Following are two types of parasites commonly encountered in the barbering environment:

- **Head lice** (**Figure 4-13**) are a type of parasite responsible for contagious diseases and conditions. One condition caused by an infestation of head lice is called **pediculosis capitis** (puh-dik-yuh-LOH-sis KAP-ih-tus).
- **Scabies** (SKAY-beez) is also a contagious skin disease and is caused by the itch mite, which burrows under the skin.

Contagious diseases and conditions caused by parasites should only be treated by a doctor. Contaminated countertops, tools, and equipment should be thoroughly cleaned and then disinfected with an EPA-registered disinfectant for the time recommended by the manufacturer or with a bleach solution for 10 minutes.

figure 4-13
Head lice.

IMMUNITY

Immunity (im-YOO-net-ee) refers to the ability of the body to destroy and resist pathogens and recognize infection. Immunity against disease can be either natural or acquired and is a sign of good health.

- **Natural immunity** (NATCH-uh-rul im-YOO-net-ee) is partly inherited and partly developed through healthy living.
- **Acquired immunity** (uh-KWY-erd ih-MYOO-nih-tee) is immunity that the body develops after overcoming a disease, through inoculation (such as flu vaccinations), or through exposure to natural allergens (such as pollen, cat dander, and ragweed).

**CAUTION**

Pathogenic bacteria, viruses, or fungi can enter the body through the following routes:

- **Skin:** broken or inflamed skin, such as a cut or a scratch, or a bruise (weakened tissue) or a rash, but not through intact skin, which is an effective barrier to infection
- **Mouth:** contaminated water, food, fingers, or objects
- **Nose:** inhaling infectious dust or droplets from a cough or sneeze
- **Eyes or ears:** organisms that reside in water that are commonly transmitted when the person is swimming
- **Genitals:** unprotected sex

The body prevents and controls infections through

- healthy, uncompromised skin—the body's first line of defense;
- body secretions, such as perspiration and digestive juices;
- white blood cells that destroy bacteria;
- antitoxins that counteract **toxins** (various poisonous substances produced by some microorganisms such as bacteria and viruses).



After reading this section, you will be able to:

- LO 4** Explain the differences between cleaning, disinfecting, and sterilizing.
- LO 5** Identify types of disinfectants and antiseptics appropriate for use in barbershops.

Proper infection control requires two steps: cleaning and then disinfecting with an appropriate EPA-registered disinfectant. When these two steps are followed correctly, virtually all pathogens of concern can be effectively eliminated.

figure 4-14
Autoclave.



figure 4-15
Unwashed hands can be swarming
with pathogens.



As stated, infection control has two steps: cleaning and disinfecting.

Remember that when you clean, you must remove all visible dirt and debris from tools, implements, and equipment by washing them with liquid soap and warm water and by using a clean and disinfected brush to scrub any grooved or hinged portions of the item.

When a surface is properly cleaned, the number of contaminants on the surface is greatly reduced. In turn, this reduces the risk of infection. The vast majority of contaminants and pathogens can be removed from the surfaces of tools and implements through proper cleaning. This is why cleaning is an important part of disinfecting tools and equipment. A surface must be properly cleaned before it can be properly disinfected. Using a disinfectant without cleaning first is like using mouthwash without brushing your teeth—it just does not work properly!

Cleaned surfaces can still harbor small amounts of pathogens, but the presence of fewer pathogens means infections are less likely to be spread (Figure 4-15). Applying antiseptics to your skin or washing your hands

with soap and water will drastically lower the number of pathogens on your hands. Proper hand cleaning requires rubbing the hands together and using liquid soap, warm running water, a nailbrush, and a clean towel (see Procedure 4-3). Do not underestimate proper cleaning and hand washing. They are the most powerful and important ways to prevent the spread of infection.

There are three ways to *clean* your tools and implements:

- Washing with soap and warm water and then scrubbing them with a clean and properly disinfected nailbrush
- Using an ultrasonic unit
- Using a cleaning solvent (e.g., clipper blade wash)

STEP 2: DISINFECTING

The second step of infection control is disinfection. Remember that disinfection is the process that eliminates most, but not necessarily all, microorganisms on nonporous surfaces. This process, however, is not effective against bacterial spores. In the barbershop, disinfection is extremely effective in controlling microorganisms on surfaces such as shears, clippers, and other **multiuse** (mul-tih-YUS) tools and equipment. A disinfectant used in the shop should carry an EPA registration number, and the label should clearly state the specific organisms the solution is effective against when used according to the manufacturer's product instructions.

Remember that disinfectants are products that destroy most bacteria (not including spores), fungi, and viruses on surfaces. Disinfectants are not for use on human skin, hair, or nails. Never use disinfectants as hand cleaners since this can cause skin irritation and **allergy** (AL-ur-jee), a reaction due to extreme sensitivity to certain foods, chemicals, or other normally harmless substances. All disinfectant manufacturers clearly state on the labels of their products that skin contact should be avoided; this means your skin as well as the client's skin. Do not put your fingers directly into any disinfecting solution. Disinfectants are pesticides and can be harmful if absorbed through the skin. If you mix a disinfectant in a container that is not labeled by the manufacturer, the container must be properly labeled with the contents and the date it was mixed. All concentrated disinfectants must be diluted exactly as instructed by the manufacturer on the product label.

While some customers who know that they have impaired immune systems will share that information with you, many will not because they are embarrassed, they do not know it is important, or they do not know that they have a compromised immune system. These people are at very high risk of infection if they come into contact with pathogens. Because you will not always know who these people are, it is important to practice proper infection control before every customer. One example is a diabetic customer whose immune system does not work effectively and who also has impaired healing. Most type 2 diabetics have been diabetic for 7 years



CAUTION

Read labels carefully. Manufacturers take great care to develop safe and highly effective products. However, when used improperly, many products that are otherwise safe can be rendered dangerous if you do not follow proper guidelines and directions exactly as the label instructs.



CAUTION

Disinfectants must be registered with the EPA. Look for an EPA registration number on the label.

prior to being diagnosed, which means that even if you ask, they may likely say “no” because they have not yet been diagnosed! Another example concerns clients on medication for conditions such as asthma, rheumatoid arthritis, and fibromyalgia—these medications are designed to dull the immune system, making these customers particularly susceptible to infection. Remember, you do not know everybody who sits in your chair, so provide the best in disinfection for everyone.

CAUTION

Improper mixing of disinfectants—to be weaker or more concentrated than the manufacturer’s instructions—can significantly reduce their effectiveness. Always add the disinfectant concentrate to the water when mixing and always follow the manufacturer’s instructions for proper dilution.

Safety glasses and gloves should be worn while mixing to avoid accidental contact with eyes and skin.

DID YOU KNOW?

The EPA has recently approved a new disinfectant that can be used in the barbershop and is available in a spray and an immersion form, as well as wipes: accelerated hydrogen peroxide (AHP). This disinfectant includes hydrogen peroxide in the formulation to increase the potency and performance of the product. AHP needs to be changed only every 14 days and is nontoxic to the skin and the environment.

Read the labels of all types of disinfectants closely. Choose the one that is most appropriate for its intended use and is the safest for you and your customers.

CHOOSING A DISINFECTANT

You must read and follow the manufacturer’s instructions whenever you are using a disinfectant. Mixing ratios (dilution) and contact time (the time as listed on the product label required for the disinfectant to be visibly moist to be effective against pathogens) are very important and can vary widely based on manufacturer and delivery method. For example, most concentrates have a 10-minute contact time, whereas most wipes have a 2-minute contact time. Not all disinfectants have the same concentration, so be sure to mix the correct proportions according to the instructions on the label. If the label does not have the word *concentrate* on it, the product is already mixed and must be used directly from the original container and must not be diluted. All EPA-registered disinfectants, even those sprayed on large surfaces, will specify a contact time in their directions for use. Disinfectants must have **efficacy** (ef-ih-KUH-see) claims on the label. Efficacy is the ability to produce the intended effect. As applied to disinfectant claims, efficacy means the effectiveness with which a disinfecting solution kills organisms when used according to the label instructions.

PROPER USE OF DISINFECTANTS

Implements must be thoroughly cleaned of all visible matter or residue before being placed in disinfectant solution. This is because residue will interfere with the disinfectant and prevent proper disinfection. Properly cleaned implements and tools, free from all visible debris, must be completely immersed in disinfectant solution. Complete immersion means there is enough liquid in the container to cover all surfaces of the item being disinfected, including the handles, for 10 minutes or for the time recommended by the manufacturer (**Figure 4-16**). When using an aerosol disinfectant, you must still look for and adhere to the contact time to ensure that all pathogens on the label are being effectively destroyed.

Disinfectant Tips

- Use disinfectants only on clean, hard, nonporous surfaces.
- Always wear gloves and safety glasses when handling disinfectant solutions.
- Always dilute products according to the instructions on the product label.
- An item must remain submerged in the disinfectant for 10 minutes unless the product label specifies differently.

- To disinfect large surfaces such as countertops, carefully apply the disinfectant onto the clean surface or use a disinfectant spray and allow it to remain moist for 10 minutes, unless the product label specifies differently.
- If the product label states “complete immersion,” the entire implement must be completely immersed in the solution.
- Change the disinfectant according to the instructions on the label. If the liquid is not changed as instructed, it will no longer be effective and may begin to promote the growth of microbes.

TYPES OF DISINFECTANTS

Disinfectants are not all the same. Some are appropriate for use in the shop and some are not. You should be aware of the different types of disinfectants and the ones that are recommended for barbershop use.

Disinfectants Appropriate for Barbershop Use

Quaternary ammonium compounds (KWAT-ur-nayr-ee uh-MOH-nee-um KAHM-powndz), also known as quats (KWATZ), are disinfectants that are very effective when used properly. The most advanced type of these formulations is called *multiple quats*. Multiple quats contain sophisticated blends of quats that work together to significantly increase the effectiveness of these disinfectants. Quat solutions usually disinfect implements in 10 minutes. These formulas may contain antirust ingredients, so leaving tools in the solution for prolonged periods can cause dulling or damage. They should be removed from the solution after the specified period, rinsed (if required), dried, and stored in a clean, covered container.

Phenolic disinfectants (fi-NOH-lik dis-in-FEK-tents) are powerful disinfectants, known as *tuberculocidal*. They are a form of formaldehyde, have a very high pH, and can damage the skin and eyes. Phenolic disinfectants can be harmful to the environment if put down the drain. They have been used reliably over the years to disinfect tools; however, they do have drawbacks. Phenol can damage plastic and rubber and can cause certain metals to rust. Extra care should be taken to avoid skin contact with phenolic disinfectants. Phenolics are known carcinogens, and as such should only be used in states that require or permit their use.

Bleach

Household bleach, 5.25 percent **sodium hypochlorite** (soh-DEE-um hy-puh-KLOR-yt), is an effective disinfectant and has been used extensively as a disinfectant in the barbershop. Using too much bleach can damage some metals and plastics, so be sure to read the label for safe use. Bleach can be corrosive to metals and plastics and can cause skin irritation and eye damage.

To mix a bleach solution, always follow the manufacturer’s directions. Store the bleach solution away from heat and light. A fresh bleach solution should be mixed every 24 hours or when the solution has been contaminated. After mixing the bleach solution, date the container to ensure that

figure 4-16
Disinfectant.



? DID YOU KNOW? Many of the aerosol products for use with clippers have several functions, such as lubricating, cooling, and rust inhibition. As long as “cleaner” and “disinfectant” are indicated on the label, the products are a “one-stop shop” for use between clients! The aerosol action does the cleaning and a disinfectant is added to the mix. Again, it is important to know the contact time for these items and to be sure to brush hair clippings from the blades before spraying.



DID YOU KNOW?

Bleach is not a magic potion!

All disinfectants, including bleach, are inactivated (made less effective) in the presence of many substances, including oils, lotions, creams, hair, and skin. If bleach is used to disinfect equipment, it is critical to use a detergent first to thoroughly clean the equipment and remove all debris. Never mix detergents with the bleach; always use bleach in a well-ventilated area.



DID YOU KNOW?

Not all household bleaches

are as effective as disinfectants. To be effective, the bleach must have an EPA registration number and contain at least 5 percent sodium hypochlorite and be diluted properly to a 10 percent solution—9 parts water to 1 part bleach.

the solution is not saved from one day to the next, but disposed of daily similar to other disinfectants. Bleach can be irritating to the lungs, so be careful about inhaling the fumes.

Petroleum Distillates

Petroleum distillates have been used by barbers for decades and are generally very similar in chemical structure to kerosene. They are excellent at removing grime and oils from metals, but not all products in this classification are approved for use as disinfectants. It is important that you check the product to ensure that it is bactericidal, virucidal, and fungicidal. The product must also be EPA registered as a disinfectant.

DISINFECTANT SAFETY

Disinfectants are pesticides (a type of poison) and can cause serious skin and eye damage. Some disinfectants appear clear while others, especially phenolic disinfectants, are a little cloudy. Always use caution when handling disinfectants, and follow the safety tips listed below.

Always

- Keep the SDS on hand for the disinfectant(s) you use.
- Wear gloves and safety glasses when mixing disinfectants.
- Avoid skin and eye contact.
- Add disinfectant to water when diluting (rather than adding water to a disinfectant) to prevent foaming, which can result in an incorrect mixing ratio.
- Use tongs, gloves, or a draining basket to remove implements from disinfectants.
- Keep disinfectants out of reach of children.
- Carefully measure and use disinfectant products according to label instructions.
- Follow the manufacturer's instructions for mixing, using, and disposing of disinfectants.
- Carefully follow the manufacturer's directions for when to replace the disinfectant solution in order to ensure the healthiest conditions for you and your client. Replace the disinfectant solution every day—more often if the solution becomes soiled or contaminated.

Never

- Let quats, phenols, bleach, or any other disinfectant come in contact with your skin. If you do get disinfectants on your skin, immediately wash the area with liquid soap and warm water. Then rinse and dry the area thoroughly.
- Place any disinfectant or other product in an unmarked container. All containers should be labeled.
- Mix chemicals together unless specified so in the manufacturer's instructions. (For example, mixing bleach and ammonia together causes toxic vapors and can cause an explosion.)

DISINFECTING CONTAINERS

In the past, jars or containers used to disinfect implements were often incorrectly called wet sanitizers. Disinfectant containers contain disinfectant for disinfecting purposes, not for cleaning. The container you choose must be large enough to contain all items to be disinfected and covered, but not airtight. Remember to clean the container every day and to wear gloves when you do. Always follow the manufacturer's label instructions for disinfecting products (Figure 4-17).

DISINFECTING NONELECTRICAL TOOLS AND IMPLEMENTS

State rules require that all multiuse tools and implements be cleaned and disinfected before and after every service—even when they are used on the same person. Mix all disinfectants according to the manufacturer's directions, always adding the disinfectant to the water, not the water to the disinfectant (see Procedure 4-1).

P 4-1 Cleaning and Disinfecting: Nonelectrical Tools and Implements
pages 86–88

DISINFECTING ELECTRICAL TOOLS AND EQUIPMENT

Hair clippers and other types of electrical equipment have contact points that cannot be completely immersed in liquid. These items should be cleaned and disinfected using an EPA-registered disinfectant designed for use on these devices. Follow the procedures recommended by the disinfectant manufacturer for preparing the solution and follow the item's manufacturer directions for cleaning and disinfecting the device (see Procedure 4-2).

P 4-2 Cleaning and Disinfecting: Clippers and Outliners pages 89–92

DISINFECTING WORK SURFACES

Before beginning a service, all work surfaces must be cleaned and disinfected. Be sure to clean and disinfect tables, stations, shampoo sinks, chairs, armrests, and any other surface that a customer's skin may have touched. Clean doorknobs and handles daily to reduce transfer of germs to your hands.

CLEANING TOWELS, LINENS, AND CAPES

Clean towels, linens, and capes must be used for each client. After a towel, linen, or cape has been used on a client, it must not be used again until it has been properly laundered. To clean towels, linens, and capes, launder according to the directions on the item's label. Be sure that towels, linens, and capes are thoroughly dried. Items that are not dry may grow mildew and bacteria. Store soiled linens and towels in covered or closed containers, away from clean linens and

figure 4-17
Disinfectants.



CAUTION

Electric sterilizers, bead sterilizers, and baby sterilizers should not be used to disinfect or sterilize implements. These devices can spread potentially infectious diseases and should never be used in the barbershop. Also, UV light units will not disinfect or sterilize implements. State rules require that you use liquid disinfecting solutions. Autoclaves are effective sterilizers. If you decide to use an autoclave, be sure that you know how to operate it properly.

figure 4-18
Strop.



towels, even if your state regulatory agency does not require that you do so. Whenever possible, use disposable towels, especially in restrooms. Do not allow the neckband of capes to touch the client's skin. All states require the use of a barrier such as, disposable neck strips or towels, to prevent the client's skin from touching the neckline of the cape.

STROPS AND LEATHER HOLSTERS

Leather is a **porous** (POHW-rus) material and as such cannot be disinfected. When utilizing a strop or holster, it is important to clean and disinfect the tool or implement after it has come into contact with the leather before using it on a client (**Figure 4-18**). It is also important that you wash your hands prior to using the strop.

Even if you are unlikely to cause a major illness, a “minor” illness spread through improper disinfection can have devastating impacts on your business. A single case of ringworm or folliculitis spread by a dirty comb or razor blade can be quickly publicized in today's social media world where bad news travels faster than good! Do not let your shop be the one posted on the web for dirty practices!

CAUTION

Products and equipment that do not have the word *disinfectant* on the label are merely cleaners. They do not disinfect. Items must be properly cleaned and disinfected after every use before using them on another client.

MULTIUSE PRODUCTS

When using creams, lotions, gels, or any other product that is dispensed from a multiuse container, it is important not to contaminate the product. Always use a pump or shaker to dispense products when possible. When products are in a tub-type container, always use a clean spatula (disposable or disinfectable) to remove the product—never use your fingers.

HAND WASHING

Properly washing your hands is one of the most important actions you can take to prevent spreading germs from one person to another. Proper hand washing removes germs from the folds and grooves of the skin and from under the free edge of the nail plate by lifting and rinsing germs and contaminants from the surface of your skin. You should wash your hands thoroughly before and after working with each customer. Follow the hand washing procedure described in Procedure 4-3.

Antimicrobial and antibacterial soaps can dry the skin, and medical studies suggest that they are no more effective than regular soaps or detergents. The true benefit of hand washing comes from the friction created by the soap bubbles that can “pull” pathogens off the skin surface. Repeated hand washing can also dry the skin, so using a moisturizing hand lotion after washing is a good practice. Be sure the hand lotion is in a pump container, not a jar.

Avoid using very hot water to wash your hands because this is another practice that can damage the skin. Remember, you must wash your hands thoroughly before and after each service, so do all you can to reduce any irritation that may occur.

P 4-3 Proper Hand Washing *pages 93–94*

WATERLESS HAND SANITIZERS

Antiseptics (ant-ih-SEP-tiks) are chemical germicides formulated for use on skin and are registered and regulated by the Food and Drug Administration. Antiseptics generally contain a high volume of alcohol and are intended to reduce the numbers and slow the growth of microbes on the skin (**Figure 4-19**). When there is visible dirt/debris on the hands, neither waterless hand sanitizers nor antiseptics will work until the dirt/debris is removed; this can only be accomplished with liquid soap, a soft-bristle brush, and water. Due to the drying effect of alcohol, hand sanitizers should not be overused, but they are an excellent option when hand washing is not possible. Never use an antiseptic to disinfect instruments or other surfaces. They are ineffective for that purpose.

CAUTION
When washing hands, use liquid soaps in pump containers. Bacteria can grow in bar soaps.

figure 4-19
Hand sanitizer.



COMMON ANTISEPTICS USED IN THE BARBERSHOP

- Hydrogen peroxide has been used in homes and barbershops virtually forever. It is generally used at 3 percent strength and works well as an antiseptic. However, it should never be used on an open cut as it destroys the cells that begin the healing process in a wound.
- Isopropyl alcohol is effective in cleaning the skin; however, it can be very drying and cause irritation of the skin. Alcohol is not a disinfectant for surfaces or implements and should only be used as a cleaner or antiseptic.

Follow Standard Precautions to Protect You and Your Clients

After reading this section, you will be able to:

- LO 6** Discuss Standard Precautions and explain procedures for handling an exposure incident.



DID YOU KNOW?

According to CDC, SP are the outgrowth of Universal Precautions (UP). UP were first introduced in 1987 to reduce the spread or the transmission of bloodborne pathogens within the healthcare setting. This initiative was designed to protect both health-care workers and patients. In 1996, UP were replaced by SP to expand the concept of UP to more situations and body fluids. In general, you should treat blood, body fluid, or tissue as though it is potentially contaminated with a communicable pathogen.



CAUTION

Taking the time to conduct a thorough hair and scalp analysis will enable you to determine whether a customer has any open wounds or abrasions. If the client does have an open wound or abrasion, do not perform services of any kind.



CAUTION

As a barber, you will work with an array of sharp implements and tools and cutting yourself is a very real possibility. If you do suffer a cut and blood is present, you must follow the steps for an exposure incident outlined in Procedure 4-4 for your safety and the safety of your client.

Standard Precautions (SP) (STAN-dard PRUH-caw-shuns) are guidelines published by the CDC that require the employer and employee to assume that any human blood and body fluids are potentially infectious. Because it may not be possible to identify customers with infectious diseases, strict infection control practices should be used with all clients. In most instances, clients who are infected with the hepatitis B virus or other bloodborne pathogens are **asymptomatic** (A-symp-toe-mat-ic), which means that they show no symptoms or signs of infection.

OSHA and CDC have set safety standards and precautions that protect employees in situations when they could be exposed to bloodborne pathogens. Precautions include proper hand washing, wearing of gloves, and proper handling and disposing of sharp instruments and any other items that may have been contaminated by blood or other body fluids. It is important that specific procedures be followed if blood or body fluid is present (see Procedure 4-4).

AN EXPOSURE INCIDENT: CONTACT WITH BLOOD OR BODY FLUID

You should never perform a service on any client who comes into the shop with an open wound, a rash, or an abrasion. Sometimes accidents happen while a service is being performed in the barbershop, however.

An **exposure incident** (eks-POH-zhoor in-SI-dent) is contact with non-intact (broken) skin, blood, body fluid, or other potentially infectious materials that is the result of the performance of a worker's duties. Should the client suffer a cut or abrasion that bleeds during a service, follow the steps outlined in Procedure 4-4 for the client's safety, as well as your own.

P 4-4 Handling an Exposure Incident *pages 95-97*

Follow Safe Work Practices and Safety Precautions

After reading this section, you will be able to:

- LO 7** Discuss safe work practices that help prevent accidents and injuries.

Most potentially harmful situations in the barbershop can be avoided by being observant and using common sense. Learn to recognize safety hazards to minimize the occurrence of accidents.